蛙声HTTP设备交互协议

1. 协议介绍

1.1 包头部分

考虑到后续可能会使用自己的私有协议，不在包头里面做过多的限制和解析内容，主要放在payload中，取消头里面与业务相关的部分

替换为Http相关的协议来做

URL:

http://192.168.0.123:8090/index/api/remote

具体的协议放在Http的body当中

- 如果没有附加数据，body类型为application/json
- 如果携带附加数据，body类型为multipart/form-data

1.2 Json部分（Http Body）

字段	类型	描述
"Seq"	int	序列号，请求端每次+1，服务端按照请求端的序列号回复
"Class"	string	跟目录的含义相同，描述一个大类，例如"Config"，"Upgrade"
"Method"	string	具体方法，例如 "Set","Get", "Upgrade"， “MultiSet”， “MultiGet”
"Param"	Json	这个格式比较灵活，根据Class.Method匹配到方法之后，由对应方法去处理对应Request/Response的参数
		对于Class.Method方法前缀为"Multi-"类型比如“MultiGet”/“MultiSet”情况下，param参数转变为数组形式，依次进行组装上传
Result"	bool	只有response才有，返回结果
"ErrorNo"	int	只有response才有，如果返回Result错误，可以设置错误码，例如参数/配置格式错误，方法不支持等

2. 举例

已经实现的协议整体上来看基本都是参数设置/获取相关的，我觉得可以理解为设置或者获取配置，随着业务的丰富，可能后续设备间或者设备与客户端之间还会存在更多种类的协同与交互，所以可以对现在参数设置与获取概括一下。例如VM33中列举的协议，可以概括成"Config.Get" "Config.Set"，如果设备端的配置采用Json,并且与设置的参数格式统一，那么协议中配置处理这一部分，几乎可以做到透明处理.

对于配置业务，现支持的方法如下

所有方法均支持多级字段，例如Encode.Preview.Audio.Chn

Method	说明	参数	说明

Set	参数配置	"Name" : "Wifi" / "Name":"Wifi.ssid"
		"Config": {
		"ssid": "this_wifi",
		"key": "12345678",
		"other": "other"
		},
		"Option":0/1
			"Param":{
			"Name":"Wifi",
			"Config":{},
			//"Options":1 可能某些配置行为需要对应选项，例如设置为配置后需要重启
			}
			新增Option字段,可以不带
			目前支持的选项值有 1：增量设置；含义：有些场景下只是修改配置中的部分字段，如果不是增量做法，就需要先获取该配置项的完整配置，再修改其中的几个字段，最后设置。增量设置支持只设置部分字段，设备端会跟原有配置进行合并，再进行设置。

Get	参数获取	"Name":"Wifi" / "Name":"Wifi.ssid"	
SetMulti	批量参数配置	Param为数组，里面是多个Name字段和Config字段	这里用数组形式， "Param":[{ "Name":"Wifi", "Config":{} }, { "Name":"OtherCfg",

			"Config":{}
			}
			]
			而不是如下形式
			"Param":{
			"Wifi" :{},
			"OtherCfg":{}
			}
			原因是，某些设置操作会有一些额外选项，这样刚好放在数组中某一项当中，Set单项中可能存在的参数或者选项，SetMulti也同样具备，避免协议导致功能退化。

GetMulti	批量参数获取	Param为数组，每一项里面都有一个"Name"	"Param":[
			{
			"Name":"Wifi"
			},
			{
			"Name":"OtherCfg"
			}
			]

~~SetDefault~~	~~设置默认参数~~	~~同Set~~	~~这个方法，是不是要对远端开放还要再看一下，感觉不是很有必要，因为默认参数在开机的时候就有了，不用等到远端过来设置~~
GetDefault	获取默认参数	同Get	同SetDefault
~~SetDefault Multi~~	~~批量设置默认参数~~	~~同SetMulti~~	~~同SetDefault~~
GetDefault Multi	批量获取默认参数	同GetMulti	同SetDefault

Restore	恢复默认参数	"Name":"Wifi"	"Param":{
			"Name":"Wifi",
			}
			不需要带配置内容，默认配置保存在设备里面

RestoreMulti	批量恢复默认参数	Param为数组	"Param":[
			{
			"Name":"Wifi"
			},
			{
			"Name":"OtherCfg"
			}
			]

具体的协议举例

设置

代码块

```
请求
{
    "Seq": 1,
    "Class": "Config",
    "Method": "Set", //"MultiSet", "Get", "MultiGet"
    "Param": {
       "Name": "Wifi",
       "Config": {
           "ssid": "this_wifi",
           "key": "12345678",
           "other": "other"
        }
     }
}
批量请求
{
    "Seq": 1,
    "Class": "Config",
    "Method": "MultiSet",
    "Param": [
        {
           "Name": "Wifi",
           "Config": {
               "ssid": "this_wifi",
               "key": "12345678",
               "other": "other"
            }
         },
         {
```

```
                "Name": "Video",
                "Config": {
                    "mode": "auto-framing"
                 }
             }
         ]
}


应答
{
    "code": 0,
    "msg": "success",
    "data": {
        "Seq": 1,
        "Class": "Config",
        "Method": "Set",
        "Param": {
            "Name": "wifi",
            "Result": true
        }
    }
}
批量应答
{
    "code":0, //成功code由200改为0
    "msg": "success",
    "data":{
        "Seq": 1,
        "Class": "Config",
        "Method": "Set",
        "Param": [
            {
                 "name":"wifi",
                 "config":{},
                 "Result":true
            }
        ]
    }
}
```

获取

代码块

```
获取
```

```
{
    "Seq": 2,
    "Class": "Config",
    "Method": "Get",
    "Param": {
        "Name": "Wifi"
    }
}
批量获取
{
    "Seq": 2,
    "Class": "Config",
    "Method": "MultiGet",
    "Param": [{
        "Name": "Wifi"
    },
    {
        "name": "Camera"
    }
    ]
}
应答
{
    "code":0, //成功code由200改为0
    "msg": "success",
    "data":{
        "Seq": 2,      //去掉了data的包裹，请求与应答保持对称的结构
        "Class": "Config",
        "Method": "MultiGet",
        "Param": {
            "Config": {
                "ssid": "this_wifi",
                "key": "12345678",
                "other": "other"
            }
        },
        "Result": true
    }
}
批量应答
{
    "code":0, //成功code由200改为0
    "msg": "success",
    "data":{
        "Seq": 2,      ////去掉了data的包裹，请求与应答保持对称的结构
        "Class": "Config",
        "Method": "MultiGet",
```

```
            "Param": [ {
                "name":"wifi"
                "Config": {
                    "ssid": "this_wifi",
                    "key": "12345678",
                    "other": "other"
                }
            },
            {
                "name":"camera"
                "Config": {
                    "ssid": "this_wifi",
                    "key": "12345678",
                    "other": "other"
                }
            }
        ]
    }
}
```

在匹配到Class.Method之后，直接取Param部分处理

Config.Set(string name, Json::Value config);

Config.Get(string name, Json::Value config);

3. wiki

3.1 http包头

http中body类型使用如下两种方式

Content-Type	说明	举例
application/json	对于大部分交互，会使用这种方式，body里面直接放一个json	
multipart/form-data	对于升级、OSD设置需要带文件或者图片的，需要使用这种方式	

3.1.1 关于multipart/form-data 说明

使用该种方式，报文的格式如下

```
POST /index/api/HamedalRemote HTTP/1.1
Content-Type: multipart/form-data; boundary=--------------------------292499742163541548715933
User-Agent: PostmanRuntime/7.29.2
Accept: */*
Postman-Token: 5a0e574f-1cc5-4514-8a34-94bc0d679b0d
Host: 192.168.16.48:8090
Accept-Encoding: gzip, deflate, br
Connection: keep-alive
Content-Length: 327518

----------------------------292499742163541548715933
Content-Disposition: form-data; name="json"

{
    "Seq": 2,
    "Class": "Config",
    "Method": "Set",
    "Param":{
        "Name":"Encode.Video.Chn1",
        "Config": {
            "BitRate": 4096,
            "BitRateControl": "CBR",
            "Chn": 1,
            "Compression": "H264",
            "Fps": 30.0,
            "GOP": 30,
            "Height": 1080,
            "PackType": 2,
            "Profile": "High",
            "Quality": 1,
            "Width": 1920
        }

    }

}
----------------------------292499742163541548715933
Content-Disposition: form-data; name="file"; filename="ota.bin"
Content-Type: application/octet-stream

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32 客户端 分组, 0 服务器 分组, 0 turn(s).
```

数据会分成多份，有一个分割符，图示中为"--------------------------292499742163541548715933"，每次请求分隔符都会变。

在body的最开始、最后、和每个数据之前都有一份分隔符。

约定如下：第一个数据分块name="json"，用来放json指令，如果是传输文件，后续的name与filename相同，都用文件名，如果只是二进制数据，则没有filename，只有name字段。

json中要放一定的字段来关联后续的二进制数据块。

3.2 Method

3.2.1 Config 配置操作

3.2.1.1 Set

Class	Method	描述	Content-Type
Config	Set	配置设置	application/json

代码块

```
请求:
{
    "Seq": 1,
    "Class": "Config",
    "Method": "Set",
    "Param": {
              "Option": 1,
          "Name": "Wifi",
               "Config": {
                        "SSID": "ssid-xxxx",
                        "key": "123456"
                }
      }
}
应答:
{
    "code": 0,
    "msg": "success",
    "data": {
        "Seq": 1,
        "Class": "Config",
        "Method": "Set",
        "Param": {
            "Name": "Wifi",
            "Result": true
        }
    }
}
```

3.2.1.2 Get

Class	Method	描述	Content-Type
Config	Get	配置获取	application/json

代码块

```
请求:
{
    "Seq": 1,
    "Class": "Config",
    "Method": "Get",
    "Param": {
        "Name": "Wifi"
    }
```

```
}

应答:
{
    "code": 0,
    "msg": "success",
    "data": {
        "Seq": 1,
        "Class": "Config",
        "Method": "Get",
        "Param": {
            "Name": "Wifi",
             "Config": {
                   "SSID": "ssid-xxxx",
                   "key": "123456"
             }
            "Result": true
        }
    }
}
```

3.2.1.3 SetMulti

Class	Method	描述	Content-Type
Config	SetMulti	设置多个配置	application/json

代码块

```
请求:
{
    "Seq": 1,
    "Class": "Config",
    "Method": "SetMulti",
    "Param": [
        {
            "Option": 1,
            "Name": "Wifi",
            "Config": {
                "SSID": "ssid-xxxx",
                "key": "123456"
            }
        },
        {
            "Option": 1,
```

```
                "Name": "LED",
                "Config": {
                    "Enable": true
                }
            }
        ]
}
应答:
{
    "code": 0,
    "msg": "success",
    "data": {
        "Seq": 1,
        "Class": "Config",
        "Method": "SetMulti",
        "Param": [
            {
                "Name": "Wifi",
                "Result": true
            },
            {
                "Name": "LED",
                "Result": true
            }
        ]
    }
}
```

3.2.1.4 GetMulti

Class	Method	描述	Content-Type
Config	GetMulti	获取多个配置	application/json

代码块

```
请求:
{
    "Seq": 1,
    "Class": "Config",
    "Method": "GetMulti",
    "Param": [
        {
            "Name": "Wifi"
        },
```

```
        {
            "Name": "LED"
        }
    ]
}
应答:
{
    "code": 0,
    "msg": "success",
    "data": {
        "Seq": 1,
        "Class": "Config",
        "Method": "GetMulti",
        "Param": [
            {
                "Name": "wifi",
                "Config": {
                    "SSID": "ssid-xxx",
                    "key": "123456"
                },
                "Result": true
            },
            {
                "Name": "LED",
                "Config": {
                    "Enable": true
                },
                "Result": true
            }
        ]
    }
```

3.2.1.5 Restore

Class	Method	描述	Content-Type
Config	Restore	配置恢复默认	application/json

代码块

```
请求:
{
    "Seq": 1,
    "Class": "Config",
    "Method": "Restore",
```

```
        "Param": {
            "Name": "Wifi"
        }
    }
    应答：
    {
        "code": 0,
        "msg": "success",
        "data": {
            "Seq": 1,
            "Class": "Config",
            "Method": "Restore",
            "Param": {
                "Name": "Wifi",
                "Result": true
            }
        }
    }
```

3.2.1.6 RestoreMulti

Class	Method	描述	Content-Type
Config	RestoreMulti	多个配置恢复默认	application/json

代码块

```
请求：
{
    "Seq": 1,
    "Class": "Config",
    "Method": "RestoreMulti",
    "Param": [
        {
            "Name": "Wifi"
        },
        {
            "Name": "LED"
        }
    ]
}
应答：
{
    "code": 0,
    "msg": "success",
```

```
        "data": {
            "Seq": 1,
            "Class": "Config",
            "Method": "RestoreMulti",
            "Param": [
                {
                    "Name": "Wifi",
                    "Result": true
                },
                {
                    "Name": "LED",
                    "Result": true
                }
            ]
        }
    }
```

3.2.1.7 RestoreAll 恢复全部默参数（设备会重启）

Class	Method	描述	Content-Type
Config	RestoreAll	恢复全部默参数	application/json

代码块

```
请求:
{
    "Seq": 1,
    "Class": "Config",
    "Method": "RestoreAll"
}
应答:
{
    "code": 0,
    "data": {
        "Class": "Config",
        "Method": "RestoreAll",
        "Param": {
            "Result": true
        },
        "Seq": 1
    },
    "msg": "success"
}
```

3.2.1.8 Subscribe 设备配置的订阅（仅对websocket有效）

Class	Method	描述	Content-Type
Config	Subscribe	配置订阅 （配置发生变化，异步通知）	application/json

代码块

```
```

```
请求，json部分
{
    "Seq": 1,
    "Class": "Config",
    "Method": "Subscribe",
    "Param":{
        "NameList":[
            {"Name":"Wifi"},
            {"Name":"DevPasswd"}
        ]
    }
}

应答:
{
    "code":0,
    "msg":"success"
}

异步通知消息:
{
    "notify”:{
        "Class":"Config",
        "Method":"Subscribe",
        "Param":{
            "Name":"Wifi",
            "Config":{
                "SSID":"ssid-xxxx",
                "key":"123456"
            }
        }
    }
}
```

3.2.1.9 UnSubscribe 取消配置订阅（仅对websocket有效）

Class	Method	描述	Content-Type
Config	UnSubscribe	取消配置订阅	application/json

代码块

```
请求，json部分
{
```

```
    "Seq": 1,
    "Class": "Config",
    "Method": "UnSubscribe",
    "Param":{
        "NameList":[
            {"Name":"Wifi"},
            {"Name":"DevPasswd"}
        ]
    }
}

应答:
{
    "code":0,
    "msg":"success"
}
```

3.2.1.10 设置AP信息

代码块

```
//请求
{
    "Seq": 1,
    "Class": "Config",
    "Method": "Set",
    "Param": {
                "Option": 1,       //必填字段，值为1
                "Name": "APInfo",
                "Config": {
                     "Ssid": "NearHub-AIR-000080",
                     "Password": "12345678"
                 }
      }
}

//响应
{
    "code": 0,
    "data": {
        "Class": "Config",
        "Method": "Set",
        "Param": {
            "Name": "APInfo",
            "Result": true,
            "Status": 0
        },
```

```
        "Seq": 1
    },
    "msg": "success"
}
```

3.2.1.11 获取AP信息

代码块

```
//请求
{
    "Seq": 1,
    "Class": "Config",
    "Method": "Get",
    "Param":{
        "Name":"APInfo"

    }
}

//响应
{
    "code": 0,
    "data": {
        "Class": "Config",
        "Method": "Get",
        "Param": {
            "Config": {
                "FreqBand": "5G",
                "Password": "12345678-123",
                "Retransmit": 1,
                "Ssid": "1-NearHub-AIR-000080"
            },
            "Name": "APInfo",
            "Result": true,
            "Status": 0
        },
        "Seq": 1
    },
    "msg": "success"
}
```

3.2.1.12 设置自动重启时间间隔

代码块

```
//请求
{
    "Seq": 1,
    "Class": "Config",
    "Method": "Set",
    "Param": {
           "Name": "RebootInterval",
           "Config": {
                "hours": 24  //  0/24/48/72/168, 设置为0表示关闭自动重启
            }
      }
}

//响应
{
    "code": 0,
    "data": {
        "Class": "Config",
        "Method": "Set",
        "Param": {
            "Name": "RebootInterval",
            "Result": true,
            "Status": 0
        },
        "Seq": 1
    },
    "msg": "success"
}
```

3.2.1.13 获取自动重启时间间隔信息

代码块

```
//请求
{
    "Seq": 1,
    "Class": "Config",
    "Method": "Get",
    "Param":{
        "Name":"RebootInterval"

    }
}

//响应
{
    "code": 0,
```

```
        "data": {
            "Class": "Config",
            "Method": "Get",
            "Param": {
                "Config": {
                    "hours": 24
                },
                "Name": "RebootInterval",
                "Result": true,
                "Status": 0
            },
            "Seq": 1
        },
        "msg": "success"
    }
```


3.2.1.14 设置Line-in音量增益

代码块


```
//请求
{
    "Seq": 1,
    "Class": "Config",
    "Method": "Set",
    "Param": {
           "Name": "Volume",
           "Config": {
                "LineIn": 2.0  // 音量数字放大系数 float类型
            }
      }
}

//响应
{
    "code": 0,
    "data": {
        "Class": "Config",
        "Method": "Set",
        "Param": {
            "Name": "Volume",
            "Result": true,
            "Status": 0
        },
        "Seq": 1
    },
    "msg": "success"
```

```
}
```

3.2.1.15 获取Line-in音量增益

代码块

```
//请求
{
    "Seq": 1,
    "Class": "Config",
    "Method": "Get",
    "Param":{
        "Name":"Volume"

    }
}

//响应
{
    "code": 0,
    "data": {
        "Class": "Config",
        "Method": "Get",
        "Param": {
            "Config": {
                "LineIn": 0.20000000000000001
            },
            "Name": "Volume",
            "Result": true,
            "Status": 0
        },
        "Seq": 1
    },
    "msg": "success"
}
```

3.2.2 Upgrade 升级

3.2.2.1 Setup

Class	Method	描述	Content-Type

Upgrade	Setup	设备升级	multipart/form-data

代码块

```
请求，json部分:
{
    "Seq": 1,
    "Class": "Upgrade",
    "Method": "Setup",
    "Session":78988798,
    "Param": {
        "Type": "Main",
        "FileName": "xxxx.bin",
        "Total": 500000,//总大小
        "Md5":"a09e878a9898dea8989" //md5,字符串长度16字节
    }
}
应答:
{
    "code": 0,
    "msg": "success",
    "data": {
        "Seq": 1,
        "Class": "Upgrade",
        "Method": "Upgrade",
        "Param": {
            "Type": "Main",
             "Result": true
        }
    }
}
```

3.2.2.2 Upgrade

Class	Method	描述	Content-Type
Upgrade	Upgrade	设备升级	multipart/form-data

代码块

```
请求，json部分:
{
    "Seq": 1,
    "Class": "Upgrade",
```

```
        "Method": "Upgrade",
        "Param": {
            "name" :"xxx.bin",
            "Pos": 2332,//当前分片索引
            "Size": 1000 //当前分片大小 建议分片大小524288
        }
    }
应答:
{
    "code": 0,
    "msg": "success",
    "data": {
        "Seq": 1,
        "Class": "Upgrade",
        "Method": "Upgrade",
        "Param": {
            "Type": "Main",
            "FileName": "xxxx.bin",
            "Total": 500000,//总大小
            "Pos": 2332,//当前分片索引
            "Size": 1000000,  //当前分片大小
             "Result": true
        }
    }
}
```

3.2.2.3 Progress

Class	Method	描述	Content-Type
Upgrade	Progress	进度查询	application/json

代码块

```
请求:
{
    "Seq": 1,
    "Class": "Upgrade",
    "Method": "Progress",
  }
应答:
{
    "code": 0,
```

```
    "msg": "success",
    "data": {
        "Seq": 1,
        "Class": "Upgrade",
        "Method": "Progress",
        "Param": {
              "Result": true
              "Progress":80
        }
    }
}
```

3.2.2.4 Version

Class	Method	描述	Content-Type
Upgrade	Version	设备版本号查询	application/json

代码块

```
请求:
{
    "Seq": 1,
    "Class": "Upgrade",
    "Method": "Version"
  }
应答:
{
    "code": 0,
    "msg": "success",
    "data": {
        "Seq": 1,
        "Class": "Upgrade",
        "Method": "Version",
        "Param": {
            "Result": true,
            "Version": "V1.100.1.xxxxxxxx"
        }
    }
}
```

3.2.2.5 云端升级

Class	Method	描述	Content-Type
Upgrade	CloudUpgrade	传输云端升级包的地址启动云端升级	application/json

代码块

```
请求:
{
    "Seq": 1,
    "Class": "Upgrade",
    "Method": "CloudUpgrade",
    "Param":{
        "Md5":"a09e878a9898dea8989",//md5,字符串长度16字节
        "Url": "https://ota-public-zh.oss-cn-shanghai.aliyuncs.com/nearhubRooms/General_MB_[Nearhub_ROOMS]_V1.0.0.0.R.230530.bin"
        "SerialNumber": "xxxxx" //设备sn/非必填
        "Destination": 2 //按位指定设备地址位2（0b10）为主设备4（0b100）为第一个子设备8（0b1000）为第二个子设备，12（0b1100）为一和二两个子设备/非必填
    }
}

应答:
{
    "code": 0,
    "msg": "success",
    "data": {
        "Seq": 1,
        "Class": "Upgrade",
        "Method": "CloudUpgrade"
    }
}

```

3.2.2.5 云端升级进度

代码块

```
请求:
{
    "Seq": 1,
    "Class": "Upgrade",
    "Method": "CloudProgress"
    "Param":{
        "serialNumber": "xxxxx" //设备sn/非必填
```

```
    }
}
应答:
{
    "code": 0,
    "data": {
        "Class": "Upgrade",
        "Method": "CloudProgress",
        "Param": {
            "schedule": 0 //100代表成功，小于0代表文件下载失败
            "serialNumber": "xxxxx" //设备sn/必填
        },
        "Seq": 1
    },
    "msg": "success"
}
```

3.2.3 KeepAlive 保活

3.2.3.1 保活

Class	Method	描述		Content-Type
KeepAlive	KeepAlive	保活		application/json

代码块

```
请求:
{
    "Seq": 1,
    "Class": "KeepAlive",
    "Method": "KeepAlive"
  }
应答:
{
    "code": 0,
    "msg": "success",
    "data": {
        "Seq": 1,
        "Class": "KeepAlive",
        "Method": "KeepAlive",
        "Param": {
              "Result": true
        }
    }
```

```
}
```

3.2.3.2 断开连接

Class	Method	描述		Content-Type
KeepAlive	UnConnect	断开连接		application/json

代码块

```
 请求:
{
    "Seq": 1,
    "Class": "KeepAlive",
    "Method": "UnConnect"
 }

 应答:
{
    "code": 0,
    "msg": "success",
    "data": {
        "Seq": 1,
        "Class": "KeepAlive",
        "Method": "UnConnect",
        "Param": {
              "Result": true
        }
    }
}
```

3.2.3.2 验证设备是否网络是否能通

代码块

```
 请求:
{
    "Ping": true
 }

 应答:
{
    "Ping": true
```

```
}
```

3.2.3 System 系统功能 校时/电量获取

3.2.3.1 SetTime

Class	Method	描述	Content-Type	说明
System	SetTime	设备校时	application/json	默认使用RFC2822格式，该格式下Type字段可以不带

代码块

```
请求:
{
    "Seq": 1,
    "Class": "System",
    "Method": "SetTime",
    "Param": {
             "Type": "RFC3339",
            "Time": "2022/10/20 14:28:20", //或"2022-10-20 14:28:20"
            "Format":"%Y/%m/%d %H:%M:%S"   //或"%Y-%m-%d %H:%M:%S"
    }
}

主要用上面这种方式


    //或者
    {
        "Param": {
        "Type": "RFC2822",
        "Format": "%a,%d %b %Y %T %Z",
        "Time": "Tue,18 Oct 2022 20:17:27 UTC"
        }
    }
    //或者
     {
        "Param": {
            "Type": "UTC", //Unix 时间戳，从1970年1月1日（UTC/GMT的午夜）开始所经过的秒数，不考虑闰秒
            "Time": 1666247438
        }
```

```
    }



应答:
{
    "code": 0,
    "msg": "success",
    "data": {
        "Seq": 1,
        "Class": "System",
        "Method": "SetTime",
        "Param": {
             "Result": true
        }
    }
}
```

3.2.3.2 SetTimeMs

Class	Method	描述	Content-Type	说明
System	SetTimeMs	设备校时	application/json	

代码块

```
请求:
{
    "Seq": 1,
    "Class": "System",
    "Method": "SetTimeMs",
    "Param": {
            "Time": 1478911019410,    //ms级utc时间, 由各平台自主获取,  *必填
            "TimeZoneOffset": 8.5,    //float类型, 支持小数的时间差,  *必填
            "TimeZoneAbbr": "ACST",   //上位机当前时区缩写,澳洲中部时区, 差8.5h, 非必填
            "TimeZoneId": "Australian"    //时区标识符, 非必填
    }
}

应答:
{
    "code": 0,
```

```
    "data": {
        "Class": "System",
        "Method": "SetTimeMs",
        "Param": {
            "Result": true,
            "Time": 1478911019436,
            "TimeZoneOffset": 8.5
        },
        "Seq": 1
    },
    "msg": "success"
}
```

3.2.3.3 GetTime

Class	Method	描述	Content-Type	说明
System	GetTime	获取设备时间	application/json	

代码块

```
请求:
{
    "Seq": 1,
    "Class": "System",
    "Method": "GetTime"
}

应答:
{
    "code": 0,
    "data": {
        "Class": "System",
        "Method": "GetTime",
        "Param": {
            "Result": true,
            "Time": 1688960886917
        },
        "Seq": 1
    },
    "msg": "success"
}
```

3.2.3.4 GetPower

Class	Method	描述	Content-Type	说明
System	GetPower	获取电量	application/json	

代码块

```
请求:
{
    "Seq": 1,
    "Class": "System",
    "Method": "GetPower"
}
应答:
{
    "code": 0,
    "data": {
        "Class": "System",
        "Method": "GetPower",
        "Param": {
            "Power": 73,
            "Result": true
        },
        "Seq": 1
    },
    "msg": "success"
}
```

3.2.3.5 SetLive

Class	Method	描述	Content-Type	说明
System	SetLive	设置直播状态	application/json	

代码块

```
请求:
{
    "Seq": 1,
    "Class": "System",
    "Method": "SetLive",
```

```
    "Param": {
        "Enable": true
    }
}
应答:
{
    "code": 0,
    "data": {
        "Class": "System",
        "Method": "SetLive",
        "Param": {
            "Result": true
        },
        "Seq": 1
    },
    "msg": "success"
}
```

3.2.3.6 CheckPasswd

Class	Method	描述	Content-Type	说明
System	CheckPasswd	校验密码	application/json	

代码块

```
请求:
{
    "Seq": 1,
    "Class": "System",
    "Method": "CheckPasswd",
    "Param": {
        "Keys": "12345678"
    }
}
应答:
{
    "code": 0,
    "data": {
        "Class": "System",
        "Method": "CheckPasswd",
        "Param": {
            "Result": true
        },
```

```
        "Seq": 1
    },
    "msg": "success"
}
```

3.2.3.7 SyncTimeOffer

Class	Method	描述	Content-Type	说明
System	SyncTimeOffer	设备校时	application/json	

代码块

```
请求:
{
    "Seq": 1,
    "Class": "System",
    "Method": "SyncTimeOffer",
    "Param": {
            "Time": 1478911019410,    //ms级utc时间，由各平台自主获取，  *必填
            "CSeq": 0,                 //每次序列号从0开始
    }
}

应答:
{
    "code": 0,
    "data": {
        "Class": "System",
        "Method": "SetTimeMs",
        "Param": {
            "Result": true,
            "Time": 1478911019410,    //ms级utc时间，由各平台自主获取，  *必填
            "CSeq": 0                 //每次序列号从0开始
        },
        "Seq": 1
    },
    "msg": "success"
}
```

3.2.3.8 SyncTimeSelect

Class	Method	描述	Content-Type	说明
System	SyncTimeSelect	设备校时	application/json	

代码块

```
请求:
{
    "Seq": 1,
    "Class": "System",
    "Method": "SyncTimeSelect",
    "Param": {
            "Time": 1478911019410,    //ms级utc时间，可选，如果存在，设备端会校验Time与CSeq对应关系
            "Delta": 10,
            "CSeq": 0                 //最终选择的序列号

            "TimeZoneOffset": 8.5,    //float类型，支持小数的时间差，  可选
    }
}

应答:
{
    "code": 0,
    "data": {
        "Class": "System",
        "Method": "SetTimeMs",
        "Param": {
            "Result": true,
            "Time": 1478911019410,    //ms级utc时间，由各平台自主获取，  *必填
            "CSeq": 0                 //最终选择的序列号
        },
        "Seq": 1
    },
    "msg": "success"
}
```

3.2.4 OSD

3.2.4.1 Set

Class	Method	描述	Content-Type	说明

OSD	Set	设置OSD	application/json multipart/form-data	两种方式都可以，如果没有图片数据都是字符串，只用json就可以，如果用字符串，就要用multi方式。 Set接口相当于覆盖方式，不会增量设置，如果要新增，用Add接口

代码块

```
JSON部分请求:
{
    "Seq": 1,
    "Class": "OSD",
    "Method": "Set",
  "Param":[
      {
        "Type":"string",
        "Content":"123456",
        "Enable":true,
        "Left" : 100,
        "Top" : 100,
        "Width": 100,
        "Height":100
    },
      {
        "Type":"bmp",
        "Content":"xxxx1.bmp",
        "Enable":true,
        "Left" : 100,
        "Top" : 100,
        "Width": 100,
        "Height":100
    },
      {
        "Type":"bmp",
        "Content":"xxxx2.bmp",
        "Enable":true,
        "Left" : 100,
        "Top" : 100,
        "Width": 100,
        "Height":100
    }
  ]
}
应答:
```

```
{
    "code": 0,
    "msg": "success",
    "data": {
        "Seq": 1,
        "Class": "OSD",
        "Method": "Set",
        "Param": {
            "Result": true
        }
    }
}
```

3.2.4.2 Add

Class	Method	描述	Content-Type	说明
OSD	Add	设置OSD	application/json multipart/form-data	两种方式都可以，如果没有图片数据都是字符串，只用json就可以，如果用字符串，就要用multi方式。 Set接口相当于覆盖方式，不会增量设置，如果要新增，用Add接口

请求应答与Set相同

3.2.5 Record 录像

3.2.5.1 Start

Class	Method	描述	Content-Type	说明
Record	Start	开始录像	application/json	Param中可以带一些参数，比如指定录像时间、录像格式等，后续开发过程中再细化

代码块

```
请求:
{
    "Seq": 1,
```

```
    "Class": "Record",
    "Method": "Start",
    "Param": {
        "RecordAll": false //默认只录制mainstream通道，该参数为true，则录制所有通道
    }
}
应答：
{
    "code": 0,
    "msg": "success",
    "data": {
        "Seq": 1,
        "Class": "Record",
        "Method": "Start",
        "Param": {
            "Result": true
        }
    }
}
```

3.2.5.2 Stop

Class	Method	描述	Content-Type	说明
Record	Stop	停止录像	application/json	

代码块

```
请求：
{
    "Seq": 1,
    "Class": "Record",
    "Method": "Stop"
}
应答：
{
    "code": 0,
    "msg": "success",
    "data": {
        "Seq": 1,
        "Class": "Record",
        "Method": "Stop",
        "Param": {
            "Result": true
```

```
        }
    }
}
```

3.2.5.3 List

Class	Method	描述	Content-Type	说明
Record	List	查询录像	application/json	Param中指定查询条件，例如时间范围，开发过程中再细化

代码块

```
请求:
{
    "Seq": 1,
    "Class": "Record",
    "Method": "List",
     "Param": {
        "Time": "2022-10-31" //以天为单位查询
    }
}
应答:
{
    "code": 0,
    "data": {
        "Class": "Record",
        "Method": "List",
        "Param": {
            "FileList": [
                "20220929_172840.mp4",
                "20220929_172939.mp4",
                "20220929_154634.mp4",
                "20220929_155114.mp4",
                "20220929_165947.mp4",
                "20220929_170412.mp4"
            ],
            "Result": true
        },
        "Seq": 1
    },
    "msg": "success"
```

```
}

```

3.2.5.4 DateList

Class	Method	描述	Content-Type	说明
Record	DateList	查询日期	application/json	

代码块

```
请求:
{
    "Seq": 1,
    "Class": "Record",
    "Method": "DateList"
}
应答:
{
    "code": 0,
    "data": {
        "Class": "Record",
        "Method": "DateList",
        "Param": {
            "DateList": [
                "2023-11-14",
                "2023-11-15",
                "2023-12-06",
                "2023-12-07"
            ],
            "Result": true
        },
        "Seq": 1
    },
    "msg": "success"
}

```

3.2.5.5 GetSd

Class	Method	描述	Content-Type	说明
Record	GetSd	获取sd卡信息	application/json	

代码块

```
请求:
{
    "Seq": 1,
    "Class": "Record",
    "Method": "GetSd"
}
应答:
{
    "code": 0,
    "data": {
        "Class": "Record",
        "Method": "GetSd",
        "Param": {
            "Capacity": 31150688,    //KB
            "Free": 31137024,        //KB
            "Path": "/mnt/sd",
            "Status": "Mount/Format/None",
            "Result": true
        },
        "Seq": 1
    },
    "msg": "success"
 }
```

3.2.5.6 FormatSd

Class	Method	描述	Content-Type	说明
Record	FormatSd	格式化sd卡	application/json	

代码块

```
请求:
{
    "Seq": 1,
    "Class": "Record",
```

```
    "Method": "FormatSd",
    "Param": {
        "Path": "/mnt/sd"
    }
}
应答:
{
    "code": 0,
    "msg": "success",
    "data": {
        "Seq": 1,
        "Class": "Record",
        "Method": "FormatSd",
        "Result": true
    }
}
```

3.2.5.7 Status

Class	Method	描述	Content-Type	说明
Record	Status	获取录制状态	application/json	

代码块

```
请求:
{
    "Seq": 1,
    "Class": "Record",
    "Method": "Status"
}
应答:
{
    "code": 0,
    "data": {
        "Class": "Record",
        "Method": "Status",
        "Param": {
            "Enable": true,
            "Result": true
        },
        "Seq": 1
    },
    "msg": "success"
```

```
}
```

3.2.5.7 livingStop

代码块

```
{
    "Seq": 1,
    "Class": "Record",
    "Method": "StopLiving"
}
```

3.2.5.8 deleteMP4

代码块

```
{
    "Seq": 1,
    "Class": "Record",
    "Method": "DeleteMP4",
     "Param": {
        "date": "2024-01-02",
        "filename": "20240102195059.mp4"
    }
}
```

3.2.6 图像聚焦

Class	Method	描述	Content-Type	说明
Focus	SetMode	聚焦模式设置	application/json	Param中可以带一些参数

3.2.6.1 SetMode

代码块

```
请求:
{
    "Seq": 1,
    "Class": "Focus",
    "Method": "SetMode",
    "Param": {
```

```
            "Mode": "Auto/Manual"
        }
}
应答:
{
    "code": 0,
    "data": {
        "Class": "Focus",
        "Method": "SetMode",
        "Param": {
            "Result": true
        },
        "Seq": 1
    },
    "msg": "success"
}
```

3.2.6.2 GetMode

Class	Method	描述	Content-Type	说明
Focus	GetMode	获取聚焦模式	application/json	Param中可以带一些参数

代码块

```
请求:
{
    "Seq": 1,
    "Class": "Focus",
    "Method": "GetMode"
}
应答:
{
    "code": 0,
    "data": {
        "Class": "Focus",
        "Method": "GetMode",
        "Param": {
            "Mode": "Auto/Manual",
            "Result": true
        },
        "Seq": 1
    },
    "msg": "success"
```

```
}
```

3.2.6.3 ManualFocus

Class	Method	描述	Content-Type	说明
Focus	Manual	手动聚焦设置	application/json	Param中可以带一些参数

代码块

```
请求:
{
    "Seq": 1,
    "Class": "Focus",
    "Method": "Manual",
    "Param": {
        "Dir": "Near/Far",
        "Speed": 100, //预留
        "Enable": true
    }
}
应答:
{
    "code": 0,
    "data": {
        "Class": "Focus",
        "Method": "Manual",
        "Param": {
            "Result": true
        },
        "Seq": 1
    },
    "msg": "success"
}

```

3.2.6.4 ManualZoom

Class	Method	描述	Content-Type	说明

Focus	ManualZoom	手动变倍设置	application/json	Param中可以带一些参数

代码块

```
请求:
{
    "Seq": 1,
    "Class": "Focus",
    "Method": "ManualZoom",
    "Param": {
        "Dir": "In/Out",
        "Speed": 100, //预留
        "Enable": true
    }
}
应答:
{
    "code": 0,
    "data": {
        "Class": "Focus",
        "Method": "ManualZoom",
        "Param": {
            "Result": true
        },
        "Seq": 1
    },
    "msg": "success"
}

```

3.2.6.5 SetZoomSpeedMode

Class	Method	描述	Content-Type	说明
Focus	SetZoomSpeedMode	设置放缩速度	application/json	Param中可以带一些参数

代码块

```
请求:
{
    "Seq": 1,
```

```
    "Class": "Focus",
    "Method": "SetZoomSpeedMode",
    "Param": {
        "SpeedMode": "Slow/Fast"
    }
}
应答:
{
    "code": 0,
    "data": {
        "Class": "Focus",
        "Method": "SetZoomSpeedMode",
        "Param": {
            "Result": true
        },
        "Seq": 1
    },
    "msg": "success"
}
```

3.2.6.6 GetZoomSpeedMode

Class	Method	描述	Content-Type	说明
Focus	GetZoomSpeedMode	获取放缩速度	application/json	Param中可以带一些参数

代码块

```
请求:
{
    "Seq": 1,
    "Class": "Focus",
    "Method": "GetZoomSpeedMode",
    "Param": {
        "SpeedMode": "Slow/Fast"
    }
}
应答:
{
    "code": 0,
    "data": {
        "Class": "Focus",
        "Method": "GetZoomSpeedMode",
        "Param": {
```

```
            "Result": true
        },
        "Seq": 1
    },
    "msg": "success"
}
```

3.2.6.7 DigitalZoom

Class	Method	描述	Content-Type	说明
Focus	DigitalZoom	数码放大	application/json	Param中可以带一些参数

代码块

```
请求:
{
    "Seq": 1,
    "Class": "Focus",
    "Method": "DigitalZoom",
    "Param": {
        "x": 1000,//放缩的中心点的横坐标
        "y": 500,//放缩的中心点的纵坐标
        "base":5.0//放缩倍率
    }
}

/*
对放缩中心点坐标和放缩倍率的限制
指定的放缩区域的左上角坐标以及区域的宽高计算如下:
rect.x = (int)(x - 1280 / base);
rect.y = (int)(y - 720 / base);
rect.w = (int)(2560 / base);
rect.h = (int)(1440 / base);

要求
rect.x >= 0
rect.y >= 0
rect.x + rect.w <= 2560
rect.y + rect.h <= 1440
base >= 1
*/

应答:
```

```
{
    "code": 0,
    "data": {
        "Class": "Focus",
        "Method": "DigitalZoom",
        "Param": {
            "Result": true
        },
        "Seq": 1
    },
    "msg": "success"
}
```

3.2.6.8 OpticsZoom

Class	Method	描述	Content-Type	说明
Focus	OpticsZoom	光学放大	application/json	Param中可以带一些参数

代码块

```
请求:
{
    "Seq": 1,
    "Class": "Focus",
    "Method": "OpticsZoom",
    "Param": {
        "base":1.0  //base范围1-8
    }
}

应答:
{
    "code": 0,
    "data": {
        "Class": "Focus",
        "Method": "OpticsZoom",
        "Param": {
            "Result": true
        },
        "Seq": 1
    },
    "msg": "success"
```

```
}
```

3.2.6.9 GetZoomInfo

Class	Method	描述	Content-Type	说明
Focus	GetZoomInfo	获取放缩信息	application/json	

代码块

```
请求:
{
    "Seq": 1,
    "Class": "Focus",
    "Method": "GetZoomInfo"
}


应答:
{
    "code": 0,
    "data": {
        "Class": "Focus",
        "Method": "GetZoomInfo",
        "Param": {
            "Result": true,
            "digitalBase": 1.0,  //数字变倍倍数
            "focusPos": 0,    //聚焦的相机位置
            "opticsBase": 1.0,  //光学变倍倍数
            "x": 1280,  //数字变倍中心点的横坐标
            "y": 720    //数字变倍中心点的纵坐标
        },
        "Seq": 1
    },
    "msg": "success"
}
```

3.2.6.10 SetFocus

Class	Method	描述	Content-Type	说明
Focus	SetFocus	手动聚焦	application/json	Param中可以带一些参数

代码块

```
请求:
{
    "Seq": 1,
    "Class": "Focus",
    "Method": "SetFocus",
    "Param": {
        "pos": 0  //镜头位置  范围为0-1478
    }
}

应答:
{
    "code": 0,
    "data": {
        "Class": "Focus",
        "Method": "SetFocus",
        "Param": {
            "Result": true
        },
        "Seq": 1
    },
    "msg": "success"
}
```

3.2.6.11 SetFocusRegion

Class	Method	描述	Content-Type	说明
Focus	SetFocusRegion	选取聚焦区域	application/json	Param中可以带一些参数; 该配置只能在focus为自动对焦模式下可以下发生效; x，y分别代表以图片左上角为原点的XY轴的比例值

代码块

```
请求:
{
    "Seq": 1,
    "Class": "Focus",
```

```
    "Method": "SetFocusRegion",
    "Param": {
        "x": 0.875,  //x轴上的比例位置，0~1
        "y": 0.426   //y轴上的比例位置，0~1
    }
}

应答：
{
    "code": 0,
    "data": {
        "Class": "Focus",
        "Method": "SetFocusRegion",
        "Param": {
            "Result": true
        },
        "Seq": 1
    },
    "msg": "success"
}
```

3.2.7 设备电源

3.2.7.1 Reboot 设备重启

Class	Method	描述	Content-Type
DevicePwr	Reboot	设备重启	application/json

代码块

```
请求：
{
    "Seq": 1,
    "Class": "DevicePwr",
    "Method": "Reboot"
    "Param": {
        "SerialNumber": "xxxxx" //设备sn/非必填
    }
}
应答：
{
    "code": 0,
    "data": {
        "Class": "DevicePwr",
```

```
            "Method": "Reboot",
            "Param": {
                "Result": true
            },
            "Seq": 1
        },
        "msg": "success"
    }
```

3.2.7.2 ShutDown 设备关机

Class	Method	描述	Content-Type
DevicePwr	ShutDown	设备关机	application/json

代码块

```
请求:
{
    "Seq": 1,
    "Class": "DevicePwr",
    "Method": "ShutDown"
}
应答:
{
    "code": 0,
    "data": {
        "Class": "DevicePwr",
        "Method": "ShutDown",
        "Param": {
            "Result": true
        },
        "Seq": 1
    },
    "msg": "success"
}
```

3.2.9 Curtain幕墙（NearHub系类设备）

3.2.9.1 GetDeviceCapacity

Class	Method	描述	Content-Type	说明

Curtain	GetDeviceCapacity	获取设备拼接能力	application/json	

代码块

```
JSON部分请求:
{
    "Seq": 1,
    "Class": "Curtain",
    "Method": "GetDeviceCapacity",
    "Param":[
      {
        "Resolution":"1080P", //分辨率
        "Caps":[
            {
                "SteramTypeDif": 1,  //不同流的种类
                "MaxPictureNum": 16 //最大支持的拼接画面数
            }
            {
                "SteramTypeDif": 2,  //不同流的种类
                "MaxPictureNum": 8  //最大支持的拼接画面数
            }
            {
                "SteramTypeDif": 3, //不同流的种类
                "MaxPictureNum": 5 //最大支持的拼接画面数
            }
            {
                "SteramTypeDif": 4,  //不同流的种类
                "MaxPictureNum": 4 //最大支持的拼接画面数
            }
        ]
      },
      {
        "Resolution":"4K", //分辨率
        "Caps":[
            ...
        ]
      },
      {
        ...
      }
    ]
}
应答:
```

```
{
    "code": 0,
    "msg": "success",
    "data": {
        "Seq": 1,
        "Class": "Curtain",
        "Method": "GetDeviceCapacity",
        "Param": {
            "Result": true
        }
    }
}
```

3.2.9.2 GetStreamType

Class	Method	描述	Content-Type	说明
Curtain	GetStreamType	获取流的类型	application/json	

代码块

```
JSON部分请求:
{
    "Seq": 1,
    "Class": "Curtain",
    "Method": "GetStreamType",
}

应答:
{
    "code": 0,
    "msg": "success",
    "data": {
        "Seq": 1,
        "Class": "Curtain",
        "Method": "GetStreamType",
        "Param":
        {
            "SupportType":     //支持流的类型
            [
                {
                    "Type": "MIRROR" //投屏，private、airplay、miracast
                    "StreamAbility": 0, //0-音视频，1-纯视频，  2-纯音频
```

```
                    "Content": "private" //辅助提示信息,当前镜像投屏类型——三种协议(private、airplay、miracast)
                },
                {
                    "Type": "USBTYPEAIN" //USBTYPEA 输入***
                    "Modules":
                    [
                        {
                            "StreamAbility": 0, //0-音视频, 1-纯视频,  2-纯音频
                            "Content": "V415&usb-1.1" //辅助提示信息
                        },
                        {
                            "StreamAbility": 2, //0-音视频, 1-纯视频,  2-纯音频
                            "Content": "A20&usb-1.2" //辅助提示信息
                        }
                    ]
                },
                {
                    "Type": "HDMIIN"
                    "StreamAbility": 0, //0-音视频, 1-纯视频,  2-纯音频
                },
                {
                    "Type": "OSD" //暂不使用
                    "StreamAbility": 1, //0-音视频, 1-纯视频,  2-纯音频
                },
                {
                    "Type": "PICTUREFILE" //SD卡背景图片
                    "Modules":
                    [
                        {
                            "StreamAbility": 1, //0-音视频, 1-纯视频,  2-纯音频
                            "Content": "111.jpg" //辅助提示信息
                        },
                        {
                            "StreamAbility": 1, //0-音视频, 1-纯视频,  2-纯音频
                            "Content": "222.jpg" //辅助提示信息
                        }
                    ]
                },
                {
                    "Type": "VIDEOFILE" //SD卡视频文件
                    "Modules":
                    [
                        {
                            "StreamAbility": 0, //0-音视频, 1-纯视频,  2-纯音频
                            "Content": "111.mp4" //辅助提示信息
                        },
```

```
                        {
                            "StreamAbility": 1, //0-音视频，1-纯视频， 2-纯音频
                            "Content": "222.mp4" //辅助提示信息
                        },
                        {
                            "StreamAbility": 2, //0-音视频，1-纯视频， 2-纯音频
                            "Content": "333.mp4" //辅助提示信息
                        },
                        {
                            "StreamAbility": 0, //0-音视频，1-纯视频， 2-纯音频
                            "Content": "444.mp4" //辅助提示信息
                        }
                    ]
                },
                {
                    "Type": "RTSP" //RTSP输入
                    "Modules":
                    [
                        {
                            "StreamAbility": 0, //0-音视频，1-纯视频， 2-纯音频
                            "Content": "v520d_090M0083&rtsp://192.168.16.30:554/live/mainstream" //辅助提示信息
                        },
                        {
                            "StreamAbility": 1, //0-音视频，1-纯视频， 2-纯音频
                            "Content": "v520d_090M0090&rtsp://192.168.16.31:554/live/mainstream" //辅助提示信息
                        }
                    ]
                } ,
                {
                    "Type": "LINEIN" //LINEIN音频输入
                    "StreamAbility": 2, //0-音视频，1-纯视频， 2-纯音频
                },
                {
                    "Type": "MIC" //MIC音频输入
                    "StreamAbility": 2, //0-音视频，1-纯视频， 2-纯音频
                } ,
                {
                    "Type": "BLUETOOTH" //蓝牙音频输入
                    "StreamAbility": 2, //0-音视频，1-纯视频， 2-纯音频
                },
                {
                    "Type": "USBTYPECIN" //USBTYPEC 输入***
                    "StreamAbility": 2, //0-音视频，1-纯视频， 2-纯音频
                }
            ]
```

```
            "Result": true
        }
    }
}
```

~~3.2.9.3 SetPlayStream~~

Class	Method	描述	Content-Type	说明
Curtain	SetPlayStream	设置播放流	application/json	上传TF卡中的图片格式JPEG,尺寸1920*1080； 文件流是视频H264和音频AAC封装的MP4； **背景图片**默认在0通道，OSD在最后通道； 总通道数不超过16； 图片、MIRROR、UVC、HDMI、OSD同时只有一路。 除了在最后新增一路流，其他操作需要停止当前播放的幕墙。

代码块

```
JSON部分请求:
{
    "Seq": 1,
    "Class": "Curtain",
    "Method": "SetPlayStream",
    "Param":[
    {
        "Channel":0, //通道
        "Type": "PICTUREFILE", //流来源类型--背景图片文件、视频文件、MIRROR、UVC、HDMI、OSD、Network等
        "ID": 0, //可选字段, TF卡中的图片ID
        "Content":"" //可选字段
    },
    {
        "Channel":1, //通道
        "Type": "VIDEOFILE", //流来源类型--图片、视频、MIRROR、UVC、HDMI、OSD、Network等
```

```
        "ID": 0 //可选字段，TF卡中的视频ID
        "Content":"" //可选字段
    },
    {
        "Channel":2, //通道
        "Type": "MIRROR", //流来源类型--图片、视频、MIRROR、UVC、HDMI、OSD、Network等
        "ID": 0 //可选字段，镜像投屏种类——三种协议(private、airplay、)
        "Content":"" //可选字段
    },
    {
        "Channel":3, //通道
        "Type": "LocalNetwork", //流来源类型--图片、视频、MIRROR、UVC、HDMI、OSD、Network等
        "ID": 0 //可选字段，LocalNetwork——拉流的设备号
        "Content":"" //可选字段
    },
    {
        "Channel":3, //通道
        "Type": "RemoteNetwork", //流来源类型--图片、视频、MIRROR、UVC、HDMI、OSD、Network等
        "ID": 0 //可不填，
        "Content": "RTSP://XXX" //可选字段，拉流的URL
    },
    {
        ...
    }
    {
        "Channel": n, //通道
        "Type": "OSD", //流来源类型--图片、视频、MIRROR、UVC、HDMI、OSD、Network等
        "Content": "欢迎光临",  //可选字段，OSD——显示字符串(后续可能要加颜色)
        "ID": 0 //可不填
    }
  ]
}
应答:
{
    "code": 0,
    "msg": "success",
    "data": {
        "Seq": 1,
        "Class": "Curtain",
        "Method": "SetPlayStream",
        "Param": {
            "Result": true
        }
    }
```

```
}
```

3.2.9.4 SetCurtainLayout

Class	Method	描述	Content-Type	说明
Curtain	SetCurtainLayout	设置场景的幕墙布局	application/json	需要拼接流时默认背景图片在底层，OSD在顶层,**层数从低到高下发**

待定参数：1. Stop是持续时间还是相对停止时间待定；

代码块

```
JSON部分请求：
{
    "Seq": 1,
    "Class": "Curtain",
    "Method": "SetCurtainLayout",
    "Param":
    {
      "Number": 0, //场景编号
      "CurtainName": "rooms1" //场景名称
      "Action" : true,//增加（true），删除（false）场景
      "DymicSwitchCurtain": true, //场景是否需要动态转换（只要有一路需要动态切就为true），true-需要，false-不需要***
      "StreamSource": //不同输出接口的音视频源配置信息
      [
          {
            "OutputType": "NETWORKOUT", //输出接口类型，NETWORKOUT，USBOUT，HDMIOUT，LINEOUT,SDOUT
            "DymicSwitchMode": 1, //0-不动态加入，1-视频动态加入拼接，2-音频动态加入混音，3-视频动态加入拼接，音频动态加入混音
            "MirrorDisplayMode": 0, //当DymicSwitchMode不为0时，表示投屏视频拼接模式或音频混音模式，
                                    //视频时：0-无模式，1-全屏展示，2-拼接， 3-画中画 最后8位代表视频
                                    //音频时：0-无模式，1-混音， 2-替换 从后9-16位表示音频（暂时未生效）
            "VideoMixer" : //可选
            {
                "Width": 1920, //拼接画面显示宽度
                "Height": 1080, //拼接画面显示高度
                "Fps": 30, //拼接画面显示帧率
                "Detail":
```

```
                [
                    {
                        "Picture":0,//画面编号，均不同
                        "Layer": 0, //所在层，可以相同 ，层数从低到高下发

                        "Stream":{
                            "Channel":0, //通道--要求按顺序分配，若完全相同的流channel应该相同
                            "Type": "PICTUREFILE", //流来源类型--背景图片文件、视频文件、MIRROR、UVC、HDMI、OSD、Network等
                            "Content":"111.jpg" //可选字段
                        },
                        "Position":{
                            "Left" : 100,
                            "Top" : 100,
                            "Width": 100,
                            "Height":100
                        },
                        "Task":{
                            "Sequence": 0, //播放序号，可以相同
                            "Start":0,  //几秒后开始
                            "Stop":-1, //几秒后停止，-1--代表一直播放
                        }
                    },
                    {
                        "Picture":1,//画面编号
                        "Layer": 1, //所在层
                        "Stream":{
                            "Channel":1,  //通道--完全相同的流channel应该相同
                            "Type": "MIRROR", //流来源类型--图片、视频、MIRROR、UVC、HDMI、OSD、Network等
                            "Content":"airplay" //可选字段
                        },
                         "Position":{
                            "Left" : 100,
                            "Top" : 100,
                            "Width": 100,
                            "Height":100
                        }
                        "Task":{
                            "Sequence": 0, //播放序号，可以相同
                            "Start":0,  //几秒后开始
                            "Stop":100, //几秒后停止，-1--代表一直播放
                        }
                    },
                    {
                        "Picture":2,//画面编号
```

```
                        "Payer": 2, //所在层
                        "Stream":{
                            "Channel":2,  //通道--完全相同的流channel应该相同
                            "Type": "VIDEOFILE", //流来源类型--背景图片文件、视频文件、MIRROR、UVC、HDMI、OSD、Network等
                            "Content":"111.mp4" //可选字段
                        },
                        "Position":{
                            "Left" : 100,
                            "Top" : 100,
                            "Width": 100,
                            "Height":100
                        }
                        "Task":{
                            "Sequence": 1, //播放序号，可以相同
                            "Start":60,  //几秒后开始
                            "Stop":100, //几秒后停止，-1--代表一直播放
                        }
                    },
                    {
                        "Picture":3,//画面编号
                        "Payer": 2, //所在层
                        "Stream":{
                            "Channel":2,  //通道--完全相同的流channel应该相同
                            "Type": "VIDEOFILE", //流来源类型--背景图片文件、视频文件、MIRROR、UVC、HDMI、OSD、Network等
                            "Content":"111.mp4" //可选字段
                        },
                        "Position":{
                            "Left" : 100,
                            "Top" : 100,
                            "Width": 100,
                            "Height":100
                        }
                        "Task":{
                            "Sequence": 1, //播放序号，可以相同
                            "Start":60,  //几秒后开始
                            "Stop":100, //几秒后停止，-1--代表一直播放
                        }
                    } ,
                    {
                        "Picture":4,//画面编号
                        "Payer": 2, //所在层
                        "Stream":{
                            "Channel":3,  //通道--完全相同的流channel应该相同
                            "Type": "RTSP", //流来源类型--背景图片文件、视频文件、MIRROR、UVC、HDMI、OSD、Network等
```

```
                        "Content":"rtsp://192.168.16.31:554/live/test" //可选字段
                    },
                    "Position":{
                        "Left" : 100,
                        "Top" : 100,
                        "Width": 100,
                        "Height":100
                    }
                    "Task":{
                        "Sequence": 1, //播放序号，可以相同
                        "Start":60,  //几秒后开始
                        "Stop":100, //几秒后停止，-1--代表一直播放
                    }
                } ,
                {
                    "Picture":5,//画面编号
                    "Payer": 2, //所在层
                    "Stream":{
                        "Channel":4,  //通道--完全相同的流channel应该相同
                        "Type": "HDMIIN", //流来源类型--背景图片文件、视频文件、MIRROR、UVC、HDMI、OSD、Network等
                    },
                    "Position":{
                        "Left" : 100,
                        "Top" : 100,
                        "Width": 100,
                        "Height":100
                    }
                    "Task":{
                        "Sequence": 1, //播放序号，可以相同
                        "Start":60,  //几秒后开始
                        "Stop":100, //几秒后停止，-1--代表一直播放
                    }
                } ,
                {
                    "Picture":6,//画面编号
                    "Payer": 3, //所在层
                    "Stream":{
                        "Channel":3,  //通道--完全相同的流channel应该相同
                        "Type": "OSD", //流来源类型--背景图片文件、视频文件、MIRROR、UVC、HDMI、OSD、Network等
                        "Content":"Welcome to nearcast" //可选字段,用户需指定
                    },
                    "Position":{
                        "Left" : 100,
                        "Top" : 100,
```

```
                            "Width": 100,
                            "Height":100
                        }
                        "Task":{
                            "Sequence": 1, //播放序号，可以相同
                            "Start":60,  //几秒后开始
                            "Stop":100, //几秒后停止，-1--代表一直播放
                        }
                    }
                ]
             }
            "AudioMixer"://可选
            {
                "SampleRate": 32000,//音频输出采样率,暂不支持
                "ChannelNum": 2,//音频输出声道数,暂不支持
                "Detail":  //目前只配置一项音频来源
                [
                    {
                        "Stream":{
                            "Channel":0,  //音频输入通道--完全相同的流channel应该相同
                            "Type": "LINEIN", //音频流来源类型--视频文件、MIRROR、UVC、HDMI、Network、UACIN、LINEIN等
                        },

                    },
                    {
                         "Stream":{
                            "Channel":1,  //音频输入通道--完全相同的流channel应该相同
                            "Type": "VIDEOFILE", //音频流来源类型--视频文件、MIRROR、UVC、HDMI、Network、UACIN、LINEIN等
                            "Content":"111.mp4" //可选字段
                        },

                    }
                ]
            }
        },
        {
            "OutputType": "USBOUT", //输出接口类型, NETWORKOUT, USBOUT, HDMIOUT, LINEOUT,SDOUT
            "DymicSwitchMode": 1, //0-不动态加入, 1-视频动态加入拼接, 2-音频动态加入混音, 3-视频动态加入拼接, 音频动态加入混音***
            "MirrorDisplayMode": 0, //投屏视频拼接模式, 0-不加入, 1-全屏展示, 2-拼接,  3-画中画***
            "VideoMixer" : //可选
```

```
            {
                "Width": 1920, //拼接画面显示宽度
                "Height": 1080, //拼接画面显示高度
                "Fps": 30, //拼接画面显示帧率
                "Detail":
                [
                     {
                        "Picture":0,//画面编号
                        "Layer": 1, //所在层
                        "Stream":{
                            "Channel":1,  //通道--不同输出接口完全相同的流channel应该相同
                            "Type": "MIRROR", //流来源类型--图片、视频、MIRROR、UVC、HDMI、OSD、Network等
                            "Content":"airplay" //可选字段
                        },
                         "Position":{
                            "Left" : 100,
                            "Top" : 100,
                            "Width": 100,
                            "Height":100
                        }
                        "Task":{
                            "Sequence": 0, //播放序号, 可以相同
                            "Start":0,  //几秒后开始
                            "Stop":-1, //几秒后停止, -1--代表一直播放
                        }
                    }
                ]
             }
            "AudioMixer":
            {
                "SampleRate": 48000, //音频输出采样率,暂不支持
                "ChannelNum": 2,//音频输出声道数,暂不支持
                "Detail":  //目前只配置一项音频来源
                [
                    {
                        "Stream":{
                            "Channel":0,  //音频输入通道--完全相同的流channel应该相同
                            "Type": "LINEIN", //音频流来源类型--视频文件、MIRROR、UVC、HDMI、Network、UACIN、LINEIN等
                        }

                    },
                    {
                         "Stream":{
```

```
                            "Channel":2,  //音频输入通道--完全相同的流channel应该相同
                            "Type": "UACIN", //音频流来源类型--视频文件、MIRROR、UVC、HDMI、Network、UACIN、LINEIN等
                        }
                    }
                ]
            }
        }
    ]
  }
}
应答:
{
    "code": 0,
    "msg": "success",
    "data": {
        "Seq": 1,
        "Class": "Curtain",
        "Method": "SetCurtainLayout",
        "Param": {
            "Result": true
        }
    }
}
```

3.2.9.5 SetPlayStrategy

Class	Method	描述	Content-Type	说明
Curtain	SetPlayStrategy	设置场景的幕墙播放策略	application/json	

代码块

```
JSON部分请求:
{
    "Seq": 1,
    "Class": "Curtain",
    "Method": "SetPlayStrategy",
    "Param": {
        "Number": 0, //场景编号，均不同
        "PlayMode": 0,   //0--实时,RealtimePlay,  1--按任务播放,PlayByTask
        "Action": 1, // 0--停止, 1-启动
```

```
        //"Time": 0, //0-立即启动，其他具体时间，还要考虑时区？？？
        //"Looped": true, //是否循环播放
    }
}
应答：
{
    "code": 0,
    "msg": "success",
    "data": {
        "Seq": 1,
        "Class": "Curtain",
        "Method": "SetPlayStrategy",
        "Param": {
            "Result": true
        }
    }
}
```

3.2.9.6 GetCurtainLayout

代码块

```
JSON部分请求：
{
    "Seq": 1,
    "Class": "Curtain",
    "Method": "GetCurtainLayout",
}
应答：
{
    "code": 0,
    "data": {
        "Class": "Curtain",
        "Method": "GetCurtainLayout",
        "Param": {
            "Curtain": {
                "Layouts": [
                    {
                        "Action": true,
                        "Number": 0,
                        "StreamSource": [
                            {
                                "AudioMixer": {
                                    "ChannelNum": 2,
                                    "Detail": [
                                        {
                                            "Stream": {
```

```
                                                "Channel": 0,
                                                "Type": "LINEIN"
                                            }
                                        }
                                    ],
                                    "SampleRate": 32000
                                },
                                "OutputType": "USBOUT"
                            }
                        ]
                    },
                    {
                        "Action": true,
                        "Number": 6,
                        "StreamSource": [
                            {
                                "OutputType": "USBOUT",
                                "VideoMixer": {
                                    "Detail": [
                                        {
                                            "Layer": 1,
                                            "Picture": 0,
                                            "Position": {
                                                "Height": 1080,
                                                "Left": 0,
                                                "Top": 0,
                                                "Width": 1920
                                            },
                                            "Stream": {
                                                "Channel": 0,
                                                "Content": "private",
                                                "Type": "MIRROR"
                                            }
                                        }
                                    ],
                                    "Fps": 30,
                                    "Height": 1080,
                                    "Width": 1920
                                }
                            },
                            {
                                "OutputType": "NETWORKOUT",
                                "VideoMixer": {
                                    "Detail": [
                                        {
                                            "Layer": 1,
                                            "Picture": 0,
```

```
                                            "Position": {
                                                "Height": 1080,
                                                "Left": 0,
                                                "Top": 0,
                                                "Width": 1920
                                            },
                                            "Stream": {
                                                "Channel": 1,
                                                "Content": "",
                                                "Type": "HDMI"
                                            }
                                        }
                                    ],
                                    "Fps": 30,
                                    "Height": 1080,
                                    "Width": 1920
                                }
                            }
                        ]
                    }
                ]
            },
            "Result": true
        },
        "Seq": 1
    },
    "msg": "success"
}
```

3.2.9.7 GetOutputInterface

Class	Method	描述	Content-Type	说明
Curtain	GetOutputInterface	获取输出接口的类型	application/json	

代码块

```
JSON部分请求:
{
    "Seq": 1,
    "Class": "Curtain",
    "Method": "GetOutputInterface",
}
```

```

应答:
{
    "code": 0,
    "msg": "success",
    "data": {
        "Seq": 1,
        "Class": "Curtain",
        "Method": "GetOutputInterface",
        "Param":
        {
            "SupportType":    //支持接口的类型
            [
                {
                    "Type": "USBTYPECOUT" //USBTYPEC输出***
                    "OutputAbility": 0, //0-音视频, 1-纯视频,  2-纯音频
                },
                {
                    "Type": "SDOUT" //SD卡
                    "OutputAbility": 0, //0-音视频, 1-纯视频,  2-纯音频
                },
                {
                    "Type": "MIRROROUT" //镜像
                    "OutputAbility": 0, //0-音视频, 1-纯视频,  2-纯音频
                },
                {
                    "Type": "HDMIOUT" //HDMI
                    "OutputAbility": 0, //0-音视频, 1-纯视频,  2-纯音频
                },
                {
                    "Type": "NETWORKOUT" //RTSP等
                    "OutputAbility": 0, //0-音视频, 1-纯视频,  2-纯音频
                },
                {
                    "Type": "BLUETOOTHOUT" //蓝牙
                    "OutputAbility": 2, //0-音视频, 1-纯视频,  2-纯音频
                },
                {
                    "Type": "LINEOUT" //LINEOUT
                    "OutputAbility": 2, //0-音视频, 1-纯视频,  2-纯音频
                },
                {
                    "Type": "USBTYPEAOUT1" //USBTYPEA 输出1***
                    "OutputAbility": 2, //0-音视频, 1-纯视频,  2-纯音频
                },
                {
                    "Type": "USBTYPEAOUT2" ///USBTYPEA 输出2***
```

```
                    "OutputAbility": 2, //0-音视频, 1-纯视频,  2-纯音频
                },
                {
                    "Type": "BYOMOUT" //BYOM输出
                    "OutputAbility": 0, //0-音视频, 1-纯视频,  2-纯音频
                }
            ]
            "Result": true
        }
    }
}
```

3.2.9.8 GetMirrorPinCode

Class	Method	描述	Content-Type	说明
Curtain	GetMirrorPinCode	获取输出接口的类型	application/json	

代码块

```
JSON部分请求:
{
    "Seq": 1,
    "Class": "Curtain",
    "Method": "GetMirrorPinCode"
}

应答:
{
    "code": 0,
    "msg": "success",
    "data": {
        "Seq": 1,
        "Class": "Curtain",
        "Method": "GetMirrorPinCode",
        "Param":
        {
            "MirrorPinCode":   "12345678"//投屏码
        }
    }
}
```

3.2.9.9 SetDeviceShareStatus

Class	Method	描述	Content-Type	说明
Curtain	SetDeviceShareStatus	设置设备的流分享状态	application/json	

代码块

```
JSON部分请求:
{
    "Seq": 1,
    "Class": "Curtain",
    "Method": "SetDeviceShareStatus",
    "Param": {
        "Status": true, //当前设备是否需要分享
        "ShareStreamUrl": "rtsp://192.168.16.100:554/live/mainstream" //当前分享设备的拉流地址
    }
}
应答:
{
    "code": 0,
    "msg": "success",
    "data": {
        "Seq": 1,
        "Class": "Curtain",
        "Method": "SetDeviceShareStatus",
        "Param": {
            "Result": true
        }
    }
}
```

3.2.9.10 CloseCurrentLayout

Class	Method	描述	Content-Type	说明
Curtain	CloseCurrentLayout	关闭当前场景，恢复默认底图	application/json	

代码块

```
JSON部分请求:
{
    "Seq": 1,
    "Class": "Curtain",
    "Method": "CloseCurrentLayout",
}
应答:
{
    "code": 0,
    "msg": "success",
    "data": {
        "Seq": 1,
        "Class": "Curtain",
        "Method": "CloseCurrentLayout",
        "Param": {
            "Result": true
        }
    }
}
```

3.2.10 Rtsp设置

3.2.10.1 Rtsp取流用户名密码设置

Class	Method	描述	Content-Type	说明
Rtsp	setRtspAuth	设置Rtsp取流用户名密码	application/json	

代码块

```
{
    "Seq": 1,
    "Class": "Rtsp",
    "Method": "setRtspAuth",
    "Param": {
        "username": "admin1111",
        "passwosrd": "admin123"
    }
}
```

3.2.11 流文件传输

代码块

```
type字段描述:
    当type字段为streamer时传输的是横幅压缩包文件，压缩包名字为image.zip; 压缩包包括一个json描述文件和若干张图片，
    描述文件名为describe。
    describe描述信息格式:
    {
        "Streamer" :
        [
                {
                        "filename" : "first.jpg",
                        "index" : 1,
                        "showtime" : 10
                },
                {
                        "filename" : "111.jpg",
                        "index" : 2,
                        "showtime" : 10
                },
                {
                        "filename" : "222.jpg",
                        "index" : 3,
                        "showtime" : 10
                }
        ],
        "relativeend" : 1000, //横幅结束播放的时间（相对时间）
        "relativestart" : 0,  //横幅开始播放的时间（相对时间）
        "timestampend" : 0,   //暂时不用
        "timestampstart" : 0  //暂时不用
    }
    其中图片的名字需要和传输图片名字一致
```

3.2.11.1 SetFileInfo

Class	Method	描述	Content-Type	说明
StreamFile	SetFileInfo	传输图片或者视频数据到设备,初始化传输文件参数（参考Upgrade）	application/json	

代码块

```
请求:
{
        "Seq": 1,
        "Class" : "StreamFile",
        "Method" : "SetFileInfo",
        "Param" :
        {
                "Filename" : "20221109_095824.mp4",
                "Md5" : "ef2153b47fe7cd69babfe1da42c3bcd1",
                "Size" : 5491499,
                "Type" : "Streamer"//type为传输文件所对应的业务，文件当作场景来源填Curtain，文件当作横幅就填Streamer
        }
}
应答:
{
       "code" : 0,
        "data" :
        {
                "Class" : "StreamFile",
                "Method" : "SetFileInfo"
        },
        "msg" : "success"
}
```

3.2.11.2 UploadStream

Class	Method	描述	Content-Type	说明
StreamFile	UploadStream	传输图片或者视频数据到设备	application/json	multipart/form-data

代码块

```
请求:
{
    "Seq": 1,
    "Class": "StreamFile",
    "Method": "UploadStream",
    "Param": {
        "Filename": "myfile",//文件名
        "Size": 524288, //"Size": 1000 //当前分片大小 建议分片大小524288
        "Pos": 522222, //当前传输的数据索引
```

```
    }
}

应答:

{
        "code" : 0,
        "data" :
        {
                "Class" : "StreamFile",
                "Method" : "UploadStream"
        },
        "msg" : "success"
}
```

3.2.11.3 GetFileInfo

Class	Method	描述	Content-Type	说明
StreamFile	GetFileInfo	获取下载文件信息	application/json	

代码块

```
请求:
{
        "Seq": 1,
        "Class" : "StreamFile",
        "Method" : "GetFileInfo",
        "Param" :
        {
            "Type" : "Streamer"//type为传输文件所对应的业务，文件当作场景来源填Curtain，文件当作横幅就填Streamer
        }
}
应答:
{
        "code" : 0,
        "data" :
        {
                "Class" : "StreamFile",
                "Method" : "GetFileInfo"
                "Param": {
                    "Filename" : "20221109_095824.mp4",
                    "Md5" : "ef2153b47fe7cd69babfe1da42c3bcd1",
```

```
                    "Size" : 5491499
                }
        },
        "msg" : "success"
}
```

3.2.11.4 DownloadStream

代码块

```
请求:
{
        "Seq": 1,
        "Class" : "StreamFile",
        "Method" : "DownloadStream"
}
响应:
请求成功直接返回文件的二进制流
```

3.2.11.5 CloudUploadStream（web端给设备上传文件,请求由web端发起）

Class	Method	描述	Content-Type	说明
StreamFile	CloudUploadStream	云端传输图片或者视频数据到设备，设备下载完了才返回,同一时刻只有一路上传。	application/json	multipart/form-data

代码块

```
请求:
{
    "Seq": 1,
    "Class": "StreamFile",
    "Method": "CloudUploadStream",
    "Param":{
        "Filename" : "20221109_095824.zip",
        "Md5":"a09e878a9898dea8989",//md5,字符串长度16字节
        "Type" : "Streamer",//type为传输文件所对应的业务，文件当作场景来源填Curtain,文件当作横幅就填Streamer
        "Url": "http://47.100.244.165:8888/img/21/21.zip"
```

```
    }
}

应答:
{
    "code": 0, //处理完了返回结果，成功为0，失败小于0
    "msg": "success",
    "data": {
        "Seq": 1,
        "Class": "StreamFile",
        "Method": "CloudUploadStream"
    }
}

```

3.2.11.6 NotiftyCloudUpdateStream（设备端更新web端文件， 请求由设备端发起）

后续版本需要考虑的问题:

1、上传文件的MD5校验， web端是否存在上传一般就可以查看横幅文件的问题（设置一个校验完毕才允许查看的问题）

2、设备如果无法连接web端，初次连接是否需要更新横幅文件

Class	Method	描述	Content-Type	说明
StreamFile	NotiftyCloudUpdateStream	通知云端有流更新	application/json	multipart/form-data

代码块

```
请求:
{
    "Seq": 1,
    "Class": "StreamFile",
    "Method": "NotiftyCloudUpdateStream",
    "Param":{
        "Type" : "Streamer",//type为传输文件所对应的业务，文件当作场景来源填Curtain，文件当作横幅就填Streamer
        "Md5"  : "ef2153b47fe7cd69babfe1da42c3bcd1"
    }
}

应答:
{
```

```
        "code": 0,
        "msg": "success",
        "data": {
            "Seq": 1,
            "Class": "StreamFile",
            "Method": "NotiftyCloudUpdateStream",
            "Param": {
                "Url" "http://47.100.244.165:8100/fileManagement/file/upload/22"
            }
        }
    }
```

3.2.12 无线网络

3.2.12.1 ScanWiFi

Class	Method	描述	Content-Type	说明
WIfi	ScanWifi	用于扫描当前可连接的WIFI信息	application/json	multipart/form-data

代码块

```
请求:
{
    "Seq": 1,
    "Class": "Wifi",
    "Method": "ScanWifi"
}

响应:
{
    "code": 0,
    "data": {
        "Class": "Wifi",
        "Method": "ScanWifi",
        "Seq": 1,
        "apList": [
            {
                "LinkQuality": 42,
                "curConnect": false,
                "enableAuth": true,
                "ssid": "TP-LINK_DE59"
            },
```

```
            {
                "LinkQuality": 37,
                "curConnect": false,
                "enableAuth": true,
                "ssid": "Auditoryworks-room3"
            },
            {
                "LinkQuality": 31,
                "curConnect": false,
                "enableAuth": true,
                "ssid": "TP-LINK_5G_test"
            },
            {
                "LinkQuality": 30,
                "curConnect": false,
                "enableAuth": true,
                "ssid": "LWX 5016"
            },
            {
                "LinkQuality": 30,
                "curConnect": false,
                "enableAuth": true,
                "ssid": "DIRECT-GFDESKTOP-4R1JES2msOP"
            },
            {
                "LinkQuality": 28,
                "curConnect": false,
                "enableAuth": true,
                "ssid": "TP-LINK_0004"
            },
            {
                "LinkQuality": 26,
                "curConnect": false,
                "enableAuth": true,
                "ssid": "ChinaNet-jnZX-5G"
            },
            {
                "LinkQuality": 26,
                "curConnect": false,
                "enableAuth": true,
                "ssid": "Auditoryworks-5G"
            },
            {
                "LinkQuality": 25,
                "curConnect": false,
                "enableAuth": true,
                "ssid": "TP-LINK_DE59"
```

```
            },
            {
                "LinkQuality": 24,
                "curConnect": false,
                "enableAuth": true,
                "ssid": "Auditoryworks-room3"
            },
            {
                "LinkQuality": 23,
                "curConnect": false,
                "enableAuth": true,
                "ssid": "test"
            },
            {
                "LinkQuality": 22,
                "curConnect": false,
                "enableAuth": true,
                "ssid": "NearHub_AP_W541"
            },
            {
                "LinkQuality": 21,
                "curConnect": false,
                "enableAuth": true,
                "ssid": "TP-LINK_test"
            },
            {
                "LinkQuality": 21,
                "curConnect": false,
                "enableAuth": true,
                "ssid": "washeng-302"
            },
            {
                "LinkQuality": 21,
                "curConnect": false,
                "enableAuth": true,
                "ssid": "washeng-302"
            },
            {
                "LinkQuality": 21,
                "curConnect": false,
                "enableAuth": true,
                "ssid": "TP-LINK_0004"
            },
            {
                "LinkQuality": 19,
                "curConnect": false,
                "enableAuth": true,
```

```
                "ssid": "insun"
            },
            {
                "LinkQuality": 19,
                "curConnect": false,
                "enableAuth": true,
                "ssid": "ChinaNet-1E72-5G"
            },
            {
                "LinkQuality": 19,
                "curConnect": false,
                "enableAuth": true,
                "ssid": "ChinaNet-jnZX"
            },
            {
                "LinkQuality": 18,
                "curConnect": false,
                "enableAuth": true,
                "ssid": "TP-LINK_AW"
            },
            {
                "LinkQuality": 18,
                "curConnect": false,
                "enableAuth": true,
                "ssid": "washeng-302"
            },
            {
                "LinkQuality": 17,
                "curConnect": false,
                "enableAuth": true,
                "ssid": "ChinaNet-1E72"
            },
            {
                "LinkQuality": 16,
                "curConnect": false,
                "enableAuth": true,
                "ssid": "WONDERLAND"
            },
            {
                "LinkQuality": 5,
                "curConnect": false,
                "enableAuth": true,
                "ssid": "NearHub_AP_5k3k"
            },
            {
                "LinkQuality": 27,
                "curConnect": false,
```

```
                "enableAuth": true,
                "ssid": "DIRECT-GFDESKTOP-4R1JES2msOP"
            }
        ]
    },
    "msg": "success"
}
```

3.2.12.2 ConnectWifi

代码块

```
请求:
{
    "Seq": 1,
    "Class": "Wifi",
    "Method": "ConnectWifi",
    "Param": {
        "SSID": "wifiName",
        "key": "123456"
    }
}
响应:
{
    "code": 0,
    "data": {
        "Class": "Wifi",
        "Method": "ConnectWifi",
        "Seq": 1
     },
    "message": "success"
}
```

3.2.12.3 OpenApService

代码块

```
请求:
{
    "Seq": 1,
    "Class": "Wifi",
    "Method": "OpenApService",
    "Param": {
        "Password": "123456", //密码，默认none
        "FreqBand": "5G", //ap频段，2.4G，5G
        "Retransmit": -1//wifi是否转发，0-不转发，1-转发，-1-维持原样
```

```
        }
}
响应:
{
    "code": 0,
    "data": {
        "Class": "Wifi",
        "Method": "openApService",
        "Seq": 1
     },
    "message": "success"
}
```

3.2.13 有线网络

3.2.13.1 设备有线网络设置

代码块

```
请求:
{
    "Seq": 1,
    "Class": "NetWork",
    "Method": "SetNetwork",
    "Param": {
        "Dhcp": false, //是妥否动态分配IP地址网关
        "IP": "192.168.16.51",//该参数为静态地址设置
        "Mask": "255.255.255.0",//该参数为静态地址设置
        "GateWay": "192.168.16.1"//该参数为静态地址设置
    }
}

响应:
{
    "code": 0,
    "data": {
        "Class": "NetWork",
        "Method": "SetIP",
        "Seq": 1
     },
    "message": "success"
}
```

3.2.13.2 设备有线网络获取

```
请求:
{
    "Seq": 1,
    "Class": "NetWork",
    "Method": "GetNetwork"
}

响应:
{
    "code": 0,
    "data": {
        "Class": "NetWork",
        "Method": "GetNetwork",
        "Seq": 1,
        "Param": {
            "Dhcp": false,
            "IP": "192.168.16.51",
            "Mask": "255.255.255.0",
            "GateWay": "192.168.16.1"
        }
     },
    "message": "success"
}
```

3.2.13.3 获取设备有线网络连接状态

代码块

```
请求:
{
    "Seq": 1,
    "Class": "NetWork",
    "Method": "GetNetworkStatus"
}

响应:
{
    "code": 0,
    "data": {
        "Class": "NetWork",
        "Method": "GetNetworkStatus",
        "Seq": 1,
        "Param": {
            "IsConnect": false //true-连接, false-未连接
        }
     },
    "message": "success"
```

```
}
```

3.2.14 会议设置

3.2.14.1 日程设置

代码块

```
请求:
{
    "Seq": 1,
    "Class": "Meeting",
    "Method": "setCalendaring",
    "Param": {
        "timestampstart": 195214563, //开始时间戳
        "timestampend": 195214563, //结束时间戳


        "relativestart": 1000000,//单位为秒
        "relativeend": 2000000,//单位为秒
        "title": "import_metting",
        "predestor": "xiaoming"
    }
}

响应:
{
    "code": 0,
    "data": {
        "Class": "Meeting",
        "Method": "setCalendaring",
        "Seq": 1
     },
    "message": "success"
}
```

3.2.14.1 日程获取

代码块

```
请求:
{
    "Seq": 1,
    "Class": "Meeting",
    "Method": "getCalendaring"
}
```

```

响应:
{
        "code": 0,
        "data": {
                "Class": "Meeting",
                "Method": "getCalendaring",
                "Seq": 1,
                "Calanders": [{
                                "timestampstart": 195214563, //开始时间戳
                                "timestampend": 195214563, //结束时间戳
                                "relativestart": 1000000, //单位为秒
                                "relativeend": 2000000, //单位为秒
                                "title": "import_metting",
                                "predestor": "xiaoming"
                        },
                        {
                                "timestampstart": 195214563, //开始时间戳
                                "timestampend": 195214563, //结束时间戳
                                "relativestart": 1000000, //单位为秒
                                "relativeend": 2000000, //单位为秒
                                "title": "import_metting",
                                "predestor": "xiaoming"
                        }
                ]
        },
        "message": "success"
}
```

3.2.15 客户端初始化

3.2.15.1 回滚设备初始化

代码块

```
请求:
{
    "Seq": 1,
    "Class": "Reset",
    "Method": "InitReset"
}
响应:
```

3.2.16 设备状态查询

代码块

```
请求:
{
    "Seq": 1,
    "Class": "System",
    "Method": "GetDevInfo",
    "Param": {
         "Keys": "12345678"
    }
}
应答:
{
    "code": 0,
    "data": {
        "Class": "System",
        "Method": "GetDevInfo",
        "Param": {
            "UvcEnable" : true, //当前uvc状态
            "Busy" : "yes", //当前是否网络拉流
                     "no",   //空闲状态
                     "isLiving", //直播推流
                     "isRecording", //本地录像
                     "NDI"   //NDI状态
        },
        "Seq": 1
    },
    "msg": "success"
}
```

3.2.17 视频状态获取

代码块

```
请求:
{
    "Seq": 1,
    "Class": "VideoMgr",
    "Method": "Status"
}

应答:
{
    "code": 0,
    "data": {
```

```
        "Class": "VideoMgr",
        "Method": "Status",
        "Param": {
        "UVC" : {
                "Status": "On",
                "Width":1280,
                "Height":720,
                "Bitrate" : 4096
                "Compression" : "YUV"

        },
        "Live" : {
                "Status" : "On",
                "Width" : 1280,
                "Height" : 720,
                "Bitrate" : 4096
                "Compression" : "H265"
        },
        "Record" : {
                "Status" : "On",
                "Width" : 1280,
                "Height" : 720,
                "Bitrate": 4096
                "Compression" : "H265"
        },
        "NDI": {
                "Status" : "On",
                "Width" : 1280,
                "Height" : 720,
                "Bitrate": 4096
                "Compression" : "H265"
        }
        },
        "Seq": 1
    },
    "msg": "success"
}

获取剩余编码能力
{
    "Seq": 1,
    "Class": "VideoMgr",
    "Method": "GetCap"
}

应答:
{
```

```
    "code": 0,
    "data": {
        "Class": "VideoMgr",
        "Method": "Status",
        "Param": {
            //
        },
        "Seq": 1
    },
    "msg": "success"
}
```

3.2.19 设备磁盘空间管理

代码块

```
获取剩余编码能力
{
    "Seq": 1,
    "Class": "DiskServer",
    "Method": "GetDiskSpace"
}

应答:
{
    "code": 0,
    "data": {
        "Class": "DiskServer",
        "Method": "GetDiskSpace",
        "Param": {
            "TotalDiskSpace": 5245588,
            "UsableDiskSpace": 55245
        },
        "Seq": 1
    },
    "msg": "success"
}
```

3.2.20 rtp文件传输

3.2.20 获取传输路径

代码块

```
请求:
{
```

```
    "Seq": 1,
    "Class": "FtpServer",
    "Method": "GetSavePath",
    "Param": {
        "operation": "DigitalSignage"//文件传输所对应的业务
    }
}
应答
{
    "code": 0,
    "data": {
        "Class": "FtpServer",
        "Method": "GetSavePath",
        "Param": {
            "SavePath": "/media/streamer"
        },
        "Seq": 1
    },
    "msg": "success"
}
```

3.2.20 校验传输文件

代码块

```
请求:
{
    "Seq": 1,
    "Class": "FtpServer",
    "Method": "TrackFile",
    "Param": {
        "operation": "DigitalSignage",
        "FileList": [
            {
                "filename": "descrtbe",
                "md5": "f5f4eec812fa85a16c28bef44d0326fa"
            },
            {
                "filename": "image1.jpg",
                "md5": "f5f4eec812fa85a16c28bef44d0326fa"
            }
        ]
    }
}
应答:
{
    "code": 0,
```

```
    "data": {
        "Class": "FtpServer",
        "Method": "TrackFile",
        "Seq": 1
    },
    "msg": "success"
}
```

3.3 配置

3.3.1 编码配置

代码块

```
"Encode": {
        "Video": {
            "Chn0": {
                "Enable" : true,
                "Chn": 0, //channel 可以和外面的Chn0不一致
                "BitRate": 4096, //码率、单位kbps
                "BitRateControl": "CBR",
                "Compression": "H264",
                "Width": 1920,
                "Height": 1080,
                "Fps": 30.0,
                "GOP": 30,
                "Profile": "High",
                "Quality": 1,
                "PackType":2
            },
            "Chn1": {
                "Enable" : true,
                "Chn": 1, //channel
                "BitRate": 4096, //码率、单位kbps
                "BitRateControl": "CBR",
                "Compression": "H264",
                "Width": 1920,
                "Height": 1080,
                "Fps": 30.0,
                "GOP": 30,
                "Profile": "High",
                "Quality": 1,
                "PackType":2
            },
            "Chn2": {
                "Chn": 2, //channel
```

```
                "BitRate": 512,
                "BitRateControl": "CBR",
                "Compression": "H264",
                "Width": 1280,
                "Height": 720,
                "Fps": 25.0,
                "GOP": 50,
                "Profile": "High",
                "Quality": 1,
                "PackType":2
            }
        },
        "Audio": {
            "Chn0": {
                "Chn": 0,
                "Enable" : true,
                "Frequency": 48000,
                "Depth": 16,
                "Tracks": 2,
                "Compression": "AAC",
                "PackTime": 16,
                "PackType":2
            }
        }
    }
```

3.3.2 图像配置

代码块

```
"ISP":{
    "Brightness" : 50, //亮度，范围0-100
    "Contrast": 50,    //对比度，范围0-100
    "Hue":50,          //色调，范围0-100
    "Saturation":"50", //情景模式，范围0-100
    "Gamma":"50",      //范围0-100
    "Mirror":true      //镜像
    "Flip":false       //翻转
    "OSDMirror":true  //OSD镜像
    "WhiteBalanceMode": "auto" "manual" (int 0:auto,1:manual)
    "WhiteBalanceTemperature":         //
}

"VideoIn":{
    "Id0" : {
        "Color":{
```

```
            "Brightness" : 50, //亮度，范围0-100
            "Contrast": 50,    //对比度，范围0-100
            "Saturation": 50,  //饱和度，范围0-100
            "Hue": 50          //色调，范围0-100
        },
        "Options":{
            "View":{
                "Mirror":true     //true 镜像
                "Flip": true,     //true 上下翻转
                "Rotate90": 0, //0-不旋转，1-顺时针90°，2-逆时针90°
            },
            "WhiteBalance": {
                "Mode": 0, //白平衡模式，0-不开启，1-自动白平衡，15-手动
                "ColorTemp": 0
            } ,
            "Exposure":{
                "Mode": 0,//曝光等级0-自动曝光，1-手动
                "Compansation": 0//期望亮度值
            },
            "AntiFlicker":0 //防闪烁模式，0-Outdoor，1-50Hz防闪烁，2-60Hz防闪烁

        }

    },
    "Id1" : {
        "Color":{
            "Brightness" : 50, //亮度，范围0-100
            "Contrast": 50,    //对比度，范围0-100
            "Hue": 50,         //色调，范围0-100
            "Saturation": 50   //情景模式，范围0-100
        },
        "View":{
            "Mirror":true
            "Flip": true,
            "Rotate90": 0, //0-不旋转，1-顺时针90°，2-逆时针90°
        },
        "WhiteBalance": {
            "Mode": 0, //白平衡模式，0-不开启，1-自动白平衡
            "ColorTemp": 0
        } ,
        "Exposure":{
            "Exposure":0,//曝光等级0-自动曝光，1..n-1手动曝光等级数，n带时间上下限的自动曝光，n+1自定义时间手动曝光（n表示支持的曝光等级数）
            "Compansation":0//期望亮度值
        },
        "AntiFlicker":0 //防闪烁模式，0-Outdoor，1-50Hz防闪烁，2-60Hz防闪烁
    }
```

```
}
```

图像配置V2 (VM33固件版本>0807)

代码块

```

"VideoIn":{
    "Id0" : {
        "Filter":{
            "Brightness" : 50, //亮度，范围0-100
            "Contrast": 50,    //对比度，范围0-100
            "Saturation": 50,  //饱和度，范围0-100
            "Hue": 50          //色调，范围0-100
            "WhiteBalance": {
                "Mode": 0, //白平衡模式，0-不开启，1-自动白平衡，15-手动
                "ColorTemp": 0
            }
        },
        "Exposure":{
            "Mode": 0, //曝光等级0-自动曝光，1-手动调整ISO和Shutter 2-自动ISO，手动调整ev+shutter 3-自动shutter，手动调整ev+ISO
            "ShutterSpeed": 30, //快门时间，传输使用分母值，0代表自动模式，{0, 3200, 2500, 2000, 1600, 1250, 1000, 800, 640, 500, 400, 320, 250, 200, 160, 125, 100, 80, 60, 50, 40, 30}
            "ISO": 125, //传感器对光的敏感程度，0代表自动模式，{0,100, 125,160,200,250,320, 400, 500, 640, 800, 1000, 1250, 1600, 2000, 2500,3200}
            "ExposureValue": 2500 // 传感器对光的敏感程度 {-200,-170,-150,-130,-100,-70.-50,-30,0, 30,50,70,100, 130,150, 170, 200}
        },
        "View":{
                "Mirror":true    //true 镜像
                "Flip": true,    //true 上下翻转
                "Rotate90": 0, //0-不旋转，1-顺时针90°，2-逆时针90°
        },
        "AntiFlicker":0 //防闪烁模式，0-Outdoor，1-50Hz防闪烁，2-60Hz防闪烁
        "HDR": 0,
        "EIS": 0
    },
    "Id1" : {
        "Filter":{
            "Brightness" : 50, //亮度，范围0-100
            "Contrast": 50,    //对比度，范围0-100
            "Saturation": 50,  //饱和度，范围0-100
            "Hue": 50          //色调，范围0-100
            "WhiteBalance": {
                "Mode": 0, //白平衡模式，0-不开启，1-自动白平衡，15-手动
```

```
                "ColorTemp": 0
            }
        },
        "Exposure":{
            "Mode": 0, //曝光等级0-自动曝光，1-手动调整ISO和Shutter 2-自动ISO，手动调整ev+shutter 3-自动shutter，手动调整ev+ISO
            "ShutterSpeed": 30, //快门时间，传输使用分母值，0代表自动模式，{0，3200，2500，2000，1600，1250，1000，800，640，500，400，320，250，200，160，125，100，80，60，50，40，30}
            "ISO": 125, //传感器对光的敏感程度，0代表自动模式，{0,100,125,160,200,250,320，400，500，640，800，1000，1250，1600，2000，2500,3200}
            "ExposureValue": 2500 // 传感器对光的敏感程度{-200,-170,-150,-130,-100,-70.-50,-30,0，30,50,70,100，130,150，170，200}
        },
        "View":{
                "Mirror":true    //true 镜像
                "Flip": true,    //true 上下翻转
                "Rotate90": 0, //0-不旋转，1-顺时针90°，2-逆时针90°
        },
        "AntiFlicker":0 //防闪烁模式，0-Outdoor，1-50Hz防闪烁，2-60Hz防闪烁
        "HDR": 0,
        "EIS": 0
    }
}
```

3.3.3 Wifi配置

代码块

```
"Wifi": {
    "SSID": "wifiName",
    "key": "123456"
}
```

3.3.4 OSD配置

代码块

```
"OSD":[
    {
        "Type":"string",
        "Content":"123456",
        "Enable":true,
        "Left" : 100,
        "Top" : 100,
        "Width": 100,
```

```
            "Height":100
        },
        {
            "Type":"bmp",
            "Content":"/mnt/sd/osd/11.bmp",
            "Enable":true,
            "Left" : 100,
            "Top" : 100,
            "Width": 100,
            "Height":100
        }
    ]
```

3.3.5 电源管理配置

代码块

```
"AutoPowerOff": {
        "Time": 900, //单位s
        "Enable": true
}
```

3.3.6 指示灯配置

代码块

```
"LedSw": {
        "Enable": true
}
```

3.3.7 按键功能配置

代码块

```
"PowerKey":{
    "Mode":"None/Record/Live/Record_Live"//取值:"Record"对应按键按下开始录像
}
```

3.3.8 时间配置

代码块

```
"Time":{
    "Format":"yyyy-MM-dd HH:mm:ss",
```

```
    "Section":"GMT"//UTC+1 UTC-1 UTC+1.5
}
```

3.3.9 音频开关

代码块

```
"audioSw":{
    "Enable": true
}
```

3.3.10 混音模式

代码块

```
"AudioMixer":{
    "Mode": "None/Mic_Line/Mic_Uac/Line_Uac"
}
```

3.3.11 设备密码设置

代码块

```
"DevPasswd":{
    "Keys": "12345678", //管理密码，默认none
    "Enable":true //使能，默认false
}
```

3.3.12 设备名称配置

代码块

```
"DevName":{
    "Name": "nearcast_admin"//设备名称，默认Nearhub_Rooms
}
```

3.3.13 设备AP信息配置(目前AP信息的获取还是走配置协议，AP的设置走3.2.12)

代码块

```
"APInfo":{
    "Password": "123456", //密码，默认none
    "FreqBand": "5G", //ap频段，2.4G，5G
```

```
    "Retransmit": 0//wifi是否转发，0-不转发，1-转发
}
```

3.3.14 Cloud token配置

代码块

```
"Token":{
    "CloudToken": "AABBCCDD", //设备接入web管理token，默认none
}
```

3.3.15 投屏设置

代码块

```
"Mirror":{
    "Protocol": "private", //投屏类型，private、airplay（苹果）、miracast（安卓），默认private(私有)
    "DisplayStyle": 0, //0-全屏展示， 1-画中画， 2-拼接
    "ActivateSecret": "azob7lW_CtRUWjG4iTDiYWw" //激活的密文信息
}
```

3.3.16 音频音量配置

代码块

```
"Volume":{
    "Volume": 50, //Lineout音量大小配置，0~100，默认50
}
```

3.3.17 URL配置

代码块

```
"URL":{
    "WebUrl": "ws://192.168.16.100:8080/live",  //WEB端访问URL配置，默认为空
    "MediaServerUrl":"192.168.16.59"  //用于录播推流与录播文件上传
}
```

3.3.18 Overlay配置

代码块

```
"Overlay":{
    "Layout": "Guide" //Overlay布局，Guide/Minimal/GuideLeft/None，默认为Guide
    "OverlayText": "IPADDR" //text的通配符信息，如：#代表空格，**{{数据库字段}}**代表显示的对应数据，内容，默认为空
    "PreviewText": "192.168.16.30" //显示的具体text信息，默认为空
    "BackgroundColor":0 //显示字体的背景颜色，默认为0
    "TextColor":0 //显示的字体的颜色，默认为0
    "UploadOverlayUrl":"http://192.168.16.200/overlay_upload",  //文件上传地址，用于Overlay图片数据上传到云端
    "DownloadOverlayUrl":"http://192.168.16.200/overlay_download"// 文件下载地址，用于从云端Overlay图片数据
}
```

3.3.19 配置密码使能

代码块

```
"DevName":{
    "Enable": false
}
```

3.3.20 房间ID

代码块

```
"Room":{
    "Id": "12345" //web端房间管理Id，默认为空
}
```

3.3.21 默认路由网卡

代码块

```
 "DefaultRoute": {
    "NetCard": "eth0"//有线传eth0，无线传wlan0
 }
```

3.3.22 音频工作模式

代码块

```
{
    "Seq": 1,
    "Class": "Config",
    "Method": "Set",
    "Param": {
        "Name": "AudioWorkMode",
        "Option":1,
        "Config": {
            "AudioWorkMode": 0  //0 语音模式; 1: 音乐（通用）模式  （默认）;  2: 只有beamforming,无降噪
        }
    }
}
```

3.3.23 音源输入切换

代码块

```
{
    "Seq": 1,
    "Class": "Config",
    "Method": "Set",
    "Param": {
        "Name": "AudioInMode",
        "Option":1,
        "Config": {
            "AudioInMode": 0  //0表示使用内置MIC, 1表示使用linein输入
        }
    }
}
```

3.3.24 音频源列表获取

代码块

```
{
    "Seq": 1,
    "Class": "System",
    "Method": "GetAudioIn"
}


{
    "code": 0,
    "data": {
        "Class": "System",
```

```
                "Method": "GetAudioIn",
                "Param": {
                    "LineIn": 0,    //0代表没有外接LineIn,1代表有外接LineIn
                    "Result": true,
                    "UacConnected": 0    //0代表没有外接Uac,1代表有外接Uac
                },
                "Seq": 1
            },
            "msg": "success"
        }
```

3.3.25 音频源详细列表获取

代码块

```
{
    "Class": "System",
    "Method": "GetAudioConfig",
    "Seq": 1
}
```

3.3.26 音频源设置

代码块

```
{
    "Seq": 1,
    "Class": "System",
    "Method": "SetAudioConfig",
    "Param": {
        "Name": "AudioConfig",
        "Option": 1,
        "Config": {
            "Id0": {
                "priority": 0,
                "name": "Built-in Mic Array",
                "available": 1,
                "isUsing": 1
            },
            "Id1": {
                "priority": 1,
                "name": "3.5mm Line In",
                "available": 1,
                "isUsing": 1
            },
            "Id2": {
```

```
                "priority": 2,
                "name": "USB Audio In",
                "available": 1,
                "isUsing": 0
            },
            "Id3": {
                "priority": 3,
                "name": "MP3",
                "available": 0,
                "isUsing": 0
            }
        }
    }
}
```

3.3.27 混音模式

代码块

```
{
    "Seq": 1,
    "Class": "Config",
    "Method": "Set",
    "Param": {
        "Name": "AudioMix",
        "Option":1,
        "Config": {
            "AudioMix": 0
        }
    }
}

{
    "Param": {
        "Option": 1,
        "Name": "AudioMix"   // 0:默认模式 1: 优先级模式 2: 混音模式
    },
    "Class": "Config",
    "Method": "Get",
    "Seq": 1
}
```

3.3.28 设备信息列表获取

代码块

```
//request
{
    "Class": "System",
    "Method": "GetDeviceList",
    "Seq": 1
}


//response
{
    "code": 0,
    "data": {
        "Class": "System",
        "Method": "GetDeviceList",
        "Seq": 1,
        "Param":{
            //设备信息列表
            "DeviceList":[
                {
                    //网络信息
                    "Network":[
                        {
                            "Name":"eth0",
                            "Addr":"192.168.16.31",
                            "Mask":"255.255.255.0",
                            "Gateway":"192.168.16.1"
                        }
                    ],
                    //设备sn码
                    "DevSN": "1234567890tsysterm",
                    //房间ID
                    "RoomId": "",
                    //设备名称
                    "DeviceNme":"12345",
                    //版本信息
                    "Version":"v1.0.0.1",
                    //设备型号
                    "DeviceModel":"Nearhub_Rooms",
                    //磁盘信息
                    "DiskInfo":[
                        {
                            "BlockName":"/dev/mmcblk2",
                            "TotalSize":60, //MB
                            "AvailableSize":20
                        },
                        {
                            "BlockName":"/dev/mmcblk2p1",
```

```
            "TotalSize":80, //GB
            "AvailableSize":20
        }
    ]
},
{
    //设备sn码
    "DevSN": "1234567890tsysterm",
    //设备名称
    "DeviceNme":"camera1",
    //版本信息
    "Version":"v1.0.0.1",
    //设备型号
    "DeviceModel":"vm33",
    //磁盘信息
    "DiskInfo":[
        {
            "BlockName":"/dev/mmcblk2",
            "TotalSize":60, //GB
            "AvailableSize":20
        },
        {
            "BlockName":"/dev/mmcblk2p1",
            "TotalSize":80, //GB
            "AvailableSize":20
        }
   ]
},

{
    //设备名称
    "DeviceNme":"AMX100",
    //设备sn码
    "DevSN": "AW02YCA9998D017281",
    //版本信息
    "Version":"v1.0.0.1",
    //设备型号
    "DeviceModel":"AMX100",
    //设备状态0离线1在线2升级中
    "DeviceState": 1
     //磁盘信息
    "DeviceList":[
        {
            //设备名称
            "DeviceNme":"AMX50",
            //设备sn码
            "DevSN": "AW02YCA9998D017282",
```

```
                            //版本信息
                            "Version":"v1.0.0.1",
                            //设备型号
                            "DeviceModel":"AMX50",

                            "DeviceState": 1
                        },
                        {
                            //设备名称
                            "DeviceNme":"AM10P",
                            //设备sn码
                            "DevSN": "AW02YCA9998D017283",
                            //版本信息
                            "Version":"v1.0.0.1",
                            //设备型号
                            "DeviceModel":"AM10P",

                            "DeviceState": 1
                        }
                   ]
                }
            ]
       }
}
```

3.3.29 设备信息列表订阅(仅对websocket有效)

代码块

```
//request
{
    "Seq": 1,
    "Class": "System",
    "Method": "Subscribe",
    "Param":{
        "NameList":[
            {"Name":"GetDeviceList"}
        ]
    }
}

//response
{
     "code":0,
```

```
    "msg":"success"
    "data": {
        "Seq": 1,
        "Class": "System",
        "Method": "Subscribe"
    }
}

//异步通知消息:
{
    "notify":{
        "Class":"System",
        "Method":"Subscribe",
        "Param":{
            "Name":"GetDeviceList",
            "DeviceList":[
                {
                    //网络信息
                    "Network":[
                        {
                            "Name":"eth0",
                            "Addr":"192.168.16.31",
                            "Mask":"255.255.255.0",
                            "Gateway":"192.168.16.1"
                        }
                    ],
                    //设备名称
                    "DeviceNme":"12345",
                    //设备的SN码
                    "DevSN" : "invalid",
                    //版本信息
                    "Version":"v1.0.0.1",
                    //设备型号
                    "DeviceModel":"Nearhub_Rooms",
                    //磁盘信息
                    "DiskInfo":[
                        {
                            "BlockName":"/dev/mmcblk2",
                            "TotalSize":60, //GB
                            "AvailableSize":20
                        },
                        {
                            "BlockName":"/dev/mmcblk2p1",
                            "TotalSize":80, //GB
                            "AvailableSize":20
                        }
                    ]
```

```
                }，
                {
                    //网络信息
                    "Network":[
                        {
                            "Name":"eth0",
                            "Addr":"192.168.16.33",
                            "Mask":"255.255.255.0",
                            "Gateway":"192.168.16.1"
                        }
                    ]，
                    //设备名称
                    "DeviceNme":"camera1",
                    //版本信息
                    "Version":"v1.0.0.1",
                    //设备型号
                    "DeviceModel":"vm33",
                    //磁盘信息
                    "DiskInfo":[
                        {
                            "BlockName":"/dev/mmcblk2",
                            "TotalSize":60, //GB
                            "AvailableSize":20
                        }，
                        {
                            "BlockName":"/dev/mmcblk2p1",
                            "TotalSize":80, //GB
                            "AvailableSize":20
                        }
                    ]
                }
            ]
        }
    }
}



```

3.3.30 直播地址读写

代码块

```
{
    "Seq": 1,
    "Class": "Config",
```

```
    "Method": "Set",
    "Param": {
        "Name": "LiveAddress",
        "Option": 1,
        "Config": {
            "LiveAddress": "xxxxxx"
        }
    }
}

{
    "Seq": 1,
    "Class": "Config",
    "Method": "Get",
    "Param": {
        "Name": "LiveAddress",
        "Option": 1
    }
}
```

3.3.31 DEVICE/HOST读写

代码块

```
{
    "Seq": 1,
    "Class": "Config",
    "Method": "Set",
    "Param": {
        "Name": "DevState",
        "Option":1,
        "Config": {
            "DevState": "device",    //device  or host
        }
    }
}

{
    "Seq": 1,
    "Class": "Config",
    "Method": "Get",
    "Param": {
        "Name": "DevState",
        "Option":1
```

```
    }
}

```

3.3.32 GPIO读写

代码块

```
写:
{
    "Seq": 1,
    "Class": "Config",
    "Method": "Set",
    "Param": {
        "Name": "DevState",
        "Option":1,
        "Config": {
            "Enable": false            //false or true  false代表GPIO为0, true代表1
        }
    }
}

回复:
{
    "code": 0,
    "data": {
        "Class": "Config",
        "Method": "Set",
        "Param": {
            "Name": "DevState",
            "Result": true,
            "Status": 0
        },
        "Seq": 1
    },
    "msg": "success"
}

读:
{
    "Seq": 1,
    "Class": "Config",
    "Method": "Get",
    "Param": {
        "Name": "DevState",
```

```
            "Option":1
        }
}

回复:
{
    "code": 0,
    "data": {
        "Class": "Config",
        "Method": "Get",
        "Param": {
            "Config": {
                "DevState": "device",
                "Enable": false
            },
            "Name": "DevState",
            "Result": true,
            "Status": 0
        },
        "Seq": 1
    },
    "msg": "success"
}

```

3.3.33 使用USB线缆开关电源模式

代码块

```
{
    "Seq": 1,
    "Class": "Config",
    "Method": "Get",
    "Param": {
        "Name": "AutoPower",
        "Option":1
    }
}

{
    "Seq": 1,
    "Class": "Config",
    "Method": "Set",
    "Param": {
```

```
            "Name": "AutoPower",
            "Option":1,
            "Config": {
                "AutoPower": true  //true;false
            }
        }
}

设备回复:
{
    "code": 0,
    "data": {
        "Class": "Config",
        "Method": "Get",
        "Param": {
            "Config": {
                "AutoPower": true
            },
            "Name": "AutoPower",
            "Result": true,
            "Status": 0
        },
        "Seq": 1
    },
    "msg": "success"
}

{
    "code": 0,
    "data": {
        "Class": "Config",
        "Method": "Set",
        "Param": {
            "Name": "AutoPower",
            "Result": true,
            "Status": 0
        },
        "Seq": 1
    },
    "msg": "success"
}
```

3.3.34 投屏和数字标牌是否开启OSD

代码块

```
{
    "Seq": 1,
    "Class": "Config",
    "Method": "Get",
    "Param": {
        "Name": "VideoShowOSD",
        "Option":1
    }
}

{
    "Seq": 1,
    "Class": "Config",
    "Method": "Set",
    "Param": {
        "Name": "VideoShowOSD",
        "Option":1,
        "Config": {
            "ScreenMirror": true  //true;false
        }
    }
}

设备回复:
{
    "code": 0,
    "data": {
        "Class": "Config",
        "Method": "Get",
        "Param": {
            "Config": {
                "ScreenMirror": true
            },
            "Name": "VideoShowOSD",
            "Result": true,
            "Status": 0
        },
        "Seq": 1
    },
    "msg": "success"
}

{
    "code": 0,
    "data": {
        "Class": "Config",
        "Method": "Set",
```

```
            "Param": {
                "Name": "VideoShowOSD",
                "Result": true,
                "Status": 0
            },
            "Seq": 1
        },
        "msg": "success"
    }
```

3.3.35 NDI状态以及视频设置

代码块

```
NDI状态设置/查询
"NDI": {
        "Enable": true,
        "AudioEnable": true,
        "NtkPanTiltSpeed": false,
        "NtkPtz": false,
        "NtkPtzZoomSpeed": false
    }


NDI格式设置

"NDIFormat" {
    "CustomBitrate":false
    "Video": {
        "BitRate": 5000, //码率、单位kbps
        "Compression": "H264",
        "Width": 1920,
        "Height": 1080,
        "Fps": 30.0,
         "BitRateControl": "CBR",
        "GOP": 30,
        "Profile": "High",
        "Quality": 1,
        "PackType":2
    }
}



//=====================================================
获取开关状态
```

```
{
    "Seq": 1,
    "Class": "Config",
    "Method": "Get",
    "Param": {
        "Option": 1,
        "Name": "NDI"
    }
}
设置开关状态
{
    "Seq": 1,
    "Class": "Config",
    "Method": "Set",
    "Param": {
        "Option": 1,
        "Name": "NDI",
        "Config": {
            "Enable": true
        }
    }
}


设置码率
{
    "Seq": 1,
    "Class": "Config",
    "Method": "Set",
    "Param": {
        "Option": 1,
        "Name": "NDIFormat",
        "Config": {
            "CustomBitrate": false,
            "Video": {
                "BitRate": 5120,
                "Compression": "H264",
                "Width": 1920,
                "Height": 1080
            }
        }
    }
}
获取码率
{
    "Seq": 1,
    "Class": "Config",
```

```
        "Method": "Get",
        "Param": {
            "Option": 1,
            "Name": "NDIFormat",
            "Config": {
                "CustomBitrate": false,
                "Video": {
                    "BitRate": 5120,
                    "Compression": "H264",
                    "Width": 1920,
                    "Height": 1080
                }
            }
        }
    }

    //=======================================================
```

3.3.36 MJPEG传输模式

代码块

```
MJPEG状态设置/查询
"MjpegMode": {
    "BulkMode" : 0/1     //0默认，1快传
    }

//=======================================================
获取传输模式
请求
{
    "Seq": 1,
    "Class": "Config",
    "Method": "Get",
    "Param": {
        "Option": 1,
        "Name": "MjpegMode"
    }
}
应答
{
    "code": 0,
    "data": {
        "Class": "Config",
        "Method": "Get",
        "Param": {
```

```
                "Config": {
                    "BulkMode": 0
                },
                "Name": "MjpegMode",
                "Result": true,
                "Status": 0
            },
            "Seq": 1
        },
        "msg": "success"
    }


    设置传输模式
    {
        "Seq": 1,
        "Class": "Config",
        "Method": "Set",
        "Param": {
            "Option": 1,
            "Name": "MjpegMode",
            "Config": {
                "BulkMode": 0
            }
        }
    }


```

3.3.37 LineIn模式切换

代码块

```
{
    "Seq": 1,
    "Class": "Config",
    "Method": "Set",
    "Param": {
        "Name": "LineInMode",
        "Option":1,
        "Config": {
            "LineInMode": 0,  // 0: mono/mic_level; 1:stereo/line_level
            "MicValue" : 10,
            "LineValue" : 10,
            "MicLevelMix": 0,   // 0:mix; 1:left; 2:right
            "MicLevelAgc": true,  // 默认true
```

```
                "MicLevelAns": true,   // 默认true
                "LineLevelAgc": false,   //默认false
                "LineLevelAns" : false   //默认false
            }
        }
}

{
    "Param": {
        "Option": 1,
        "Name": "LineInMode"   // 0:MIC(默认)  1: LineIn
    },
    "Class": "Config",
    "Method": "Get",
    "Seq": 1
}
```

3.3.38 AgcMode模式切换

代码块

```
{
    "Seq": 1,
    "Class": "Config",
    "Method": "Set",
    "Param": {
        "Name": "AgcMode",
        "Option":1,
        "Config": {
            "AgcMode": 0,
            "NoiseReduction": 0
        }
    }
}

{
    "Param": {
        "Option": 1,
        "Name": "AgcMode"   // 0:关(默认)  1: 开
    },
    "Class": "Config",
    "Method": "Get",
    "Seq": 1
}
```

3.4 直播推流

3.4.1 开始推流

代码块

```
//请求:
{
    "Seq": 1,
    "Class": "PushStream",
    "Method": "Start",
    "Param": {
        "pushUrl": "rtsp://192.168.1.100/live-roomid/mainstream?room_id=1234",
        "channel": "mainstream" //暂时只支持mainstream通道
    }
}

//响应
{
    "code": 0,
    "data": {
        "Class": "PushStream",
        "Method": "start",
        "Seq": 1
     },
    "message": "success"
}
```

3.4.2 停止推流

代码块

```
//请求:
{
    "Seq": 1,
    "Class": "PushStream",
    "Method": "Stop",
    "Param": {
        "pushUrl": "rtsp://192.168.1.100/live-roomid/mainstream?room_id=1234"//没有该字段则停止所有推流
    }
}

```

```
//响应
{
    "code": 0,
    "data": {
        "Class": "PushStream",
        "Method": "Stop",
        "Seq": 1
     },
    "message": "success"
}
```

3.5 设备系统状态

代码块

```
//request
{
    "Seq": 1,
    "Class": "DeviceStatus",
    "Method": "Get",
}


//response
{
    "code": 0,
    "data": {
        "Class": "DeviceStatus",
        "Method": "Get",
        "Seq": 1,
        "Param":{
            //cpu 使用百分比 0~100
            "Cpu":50,
            //内存使用情况
            "Mem":{
                "total":1024, //单位MB
                "free":300
            }
            //存储sd卡使用情况
            "SDStorage":{
                "total":50, //单位GB
                "free":20
            }
        }
```

```
    },
    "message": "success"
}
```

3.6 输入源管理（rooms）

3.6.1 Add 添加输入源

代码块

```
//request
{
    "Seq": 1,
    "Class": "InputSource",
    "Method": "Add",
    "Param":{
        "Type":"NETWORK",  //NETWORK 暂时只支持网络类型,后续扩展其他类型
        "Ability": "video_audio", //video_audio-音视频, video-纯视频,  audio-纯音频
        "Content":"rtsp://192.168.16.11/live/mainstream",
        "Desc":"teacher" //输入源描述
    }
}

//response
{
    "code": 0,
    "msg": "success",
    "data":{
        "Seq": 1,
        "Class": "InputSource",
        "Method": "Add",
        "Param":{
            "SourceId":100 //
        }
    }
}

```

3.6.2 Delete 删除输入源

```
//request
{
    "Seq": 1,
    "Class": "InputSource",
    "Method": "Delete",
    "Param":{
        "SourceId":100
    }
}

//response
{
    "code": 0,
    "msg": "success",
    "data":{
        "Seq": 1,
        "Class": "InputSource",
        "Method": "Delete",
        "Param":{
            "SourceId":100 //
        }
    }
}
```

3.6.3 Set 修改输入源

代码块

```
//request
{
    "Seq": 1,
    "Class": "InputSource",
    "Method": "Set",
    "Param":{
        "SourceId":100,
        "Type":"NETWORK",  //NETWORK 暂时只支持网络类型,后续扩展其他类型
        "Ability": "video_audio", //video_audio-音视频, video-纯视频,  audio-纯音频
        "Content":"rtsp://192.168.16.11/live/mainstream",
        "Desc":"teacher" //输入源描述
    }
}

//response
{
    "code": 0,
```

```
    "msg": "success",
    "data":{
        "Seq": 1,
        "Class": "InputSource",
        "Method": "Set",
        "Param":{
            "Type":"NETWORK",   //NETWORK 暂时只支持网络类型,后续扩展其他类型
            "SourceId":100,
            "Content":"rtsp://192.168.16.11/live/mainstream"
        }
    }
}
```

3.6.4 Get 获取输入源

代码块

```
//request
{
    "Seq": 1,
    "Class": "InputSource",
    "Method": "Get",
    "Param":{
        //"SourceId":100, //无该字段，则获取所有输入源
    }
}

//response
{
    "code": 0,
    "msg": "success",
    "data":{
        "Seq": 1,
        "Class": "InputSource",
        "Method": "Get",
        "Param":{
            "InputSourceList":[
                {
                    "Type":"NETWORK",
                    "SourceId":100,
                    "Ability": "video_audio", //video_audio-音视频，video-纯视频， audio-纯音频
                    "Content":"rtsp://192.168.16.100/live/mainstream",
                    "Desc":"teacher1" //输入源描述
                },
                {
```

```
                    "Type":"NETWORK",
                    "SourceId":101,
                    "Ability": "video_audio", //video_audio-音视频, video-纯视频, audio-纯音频
                    "Content":"rtsp://192.168.16.101/live/mainstream",
                    "Desc":"teacher2" //输入源描述
                }
            ]
        }
    }
}
```

3.7 产测协议

3.7.1 Decoder产测

3.7.1.1 wifi连接

代码块

```
//request
{
    "Seq": 1,
    "Class": "ProductTest",
    "Method": "WifiConnect",
    "Param": {
        "Wifi": {
        "SSID": "wifiName",
        "Key": "123456"
        }
    }
}

//response
{
    "code": 0,
    "data": {
        "Class": "ProductTest",
        "Method": "WifiConnect",
        "Seq": 1
     },
    "message": "success"
```

```
}
```

3.7.1.2 获取wifi连接信号强度

代码块

```
//request
{
    "Seq": 1,
    "Class": "ProductTest",
    "Method": "WifiGetSignalStrength"
}

//response
{
    "code": 0,
    "data": {
        "Class": "ProductTest",
        "Method": "WifiGetSignalStrength",
        "Seq": 1,
        "Param": {
            "SignalStrength":100 //0~100
        }

     },
    "message": "success"

}
```

3.7.1.3 视频文件解码播放测试

视频文件需要提前存放在U盘分区

代码块

```
//request
{
    "Seq": 1,
    "Class": "ProductTest",
    "Method": "VideoPlay"
}

//response
{
```

```
    "code": 0,
    "data": {
        "Class": "ProductTest",
        "Method": "VideoPlay",
        "Seq": 1·
    },
    "message": "success"
}
```

3.7.1.4 Line In 与 Line Out接口测试

获取LINE IN 与 LINE OUT接口连接状态状态

代码块

```
//request
{
    "Seq": 1,
    "Class": "ProductTest",
    "Method": "GetLineStatus"
}

//response
{
    "code": 0,
    "data": {
        "Class": "ProductTest",
        "Method": "GetLineStatus",
        "Seq": 1,
        "Param": {
            "LineInStatus":true,
            "LineOutStatus":false
        }
    },
    "message": "success"
}
```

音频输入输出检测

代码块

```
//request
{
    "Seq": 1,
```

```
    "Class": "ProductTest",
    "Method": "AudioPlay"
}

//response
{
    "code": 0,
    "data": {
        "Class": "ProductTest",
        "Method": "AudioPlay",
        "Seq": 1
     },
    "message": "success"
}

```

3.7.1.5 Usb U盘接口测试

//设备检测U盘挂载，并自动做读写测试，返回结果

代码块

```
//request
{
    "Seq": 1,
    "Class": "ProductTest",
    "Method": "UsbStorage"
}

//response
{
    "code": 0,
    "data": {
        "Class": "ProductTest",
        "Method": "UsbStorage",
        "Seq": 1
     },
    "message": "success"
}
```

3.7.1.6 Kvm 接口测试

//等待键盘事件（t按键）

代码块

```
//request
{
    "Seq": 1,
    "Class": "ProductTest",
    "Method": "Kvm"
}

//response
{
    "code": 0,
    "data": {
        "Class": "ProductTest",
        "Method": "Kvm",
        "Seq": 1
     },
    "message": "success"
}
```

3.7.1.7 设置SN码

代码块

```
//request
{
    "Seq": 1,
    "Class": "ProductTest",
    "Method": "SetSN",
    "Param": {
        "SN":"1234abcfddddfd"
    }

}

//response
{
    "code": 0,
    "data": {
        "Class": "ProductTest",
        "Method": "SetSN",
        "Seq": 1
     },
```

```
    "message": "success"
}
```

3.7.1.8 获取SN码

代码块

```
//request
{
    "Seq": 1,
    "Class": "ProductTest",
    "Method": "GetSN"
}

// success response
{
    "code": 0,
    "data": {
        "Class": "ProductTest",
        "Method": "GetSN",
        "Seq": 1,
        "Param": {
            "SN":"1234abcfddddfd" //error is Invaild
            // "SN":"Invaild"
        }
     },
    "message": "success"
}

```

3.7.1.9 设置MAC地址

代码块

```
//request
{
    "Seq": 1,
    "Class": "ProductTest",
    "Method": "SetMAC",
    "Param": {
        "MAC":"ab:ff:12:34:bb:aa"
    }

}

//response
```

```
{
    "code": 0,
    "data": {
        "Class": "ProductTest",
        "Method": "SetMAC",
        "Seq": 1
     },
    "message": "success"
}
```

3.7.1.10 获取MAC地址

代码块

```
//request
{
    "Seq": 1,
    "Class": "ProductTest",
    "Method": "GetMAC"
}

//response
{
    "code": 0,
    "data": {
        "Class": "ProductTest",
        "Method": "GetMAC",
        "Seq": 1,
        "Param": {
            "MAC":"ab:ff:12:34:bb:aa"
            //"MAC":"Invaild"
        }
     },
    "message": "success"
}
```

3.7.1.11 复位按键测试

代码块

```
//request
{
    "Seq": 1,
    "Class": "ProductTest",
    "Method": "ResetButton"
}
```

```

//response
{
    "code": 0,
    "data": {
        "Class": "ProductTest",
        "Method": "ResetButton",
        "Seq": 1
     },
    "message": "success"
}
```

3.8 失败案例

代码块

```
1、失败错误码:  -600
含义:
    表示操作失败，可能是传入的参数有问题，也可能是设备端处理失败，以WIFI连接操作为例，当WIFI连接失败返回-600。
例子:
{
    "code": -600,
    "data": {
        "Class": "Wifi",
        "Method": "ConnectWifi",
        "Seq": 1
    },
    "msg": "OptFailed"
}

2、失败错误码: -500
含义:
    设备找不到对应的处理方法，大概率元原因是class或者method对不上
例子:
{
    "code": -500,
    "data": {
        "Class": "Wifi",
        "Method": "ConnectWif",
        "Seq": 1
    },
    "msg": "NotFound"
```

```
}
```

3.9 describe文件信息

3.9.1 数字标牌describe文件信息

代码块

```
{
    //数字标牌列表
    "Streamer": [
        {
            "filename":
"http://costorage.auditoryworks.co/dynamic/01HDGBTXG5W46ZPJQ4N29V1F7J/C50_home.
mp4/",//如果是在线视频指的是url, 本地文件指的时文件名
            "index": 1,//播放顺序
            "showtime": 2147483647,//播放时长
            "vpnaddr": ""//vpn
        },
                {
            "filename":
"http://costorage.auditoryworks.co/dynamic/01HDGBTXG5W46ZPJQ4N29V1F7J/V403_2.mp
4",
            "index": 2,
            "showtime": 2147483647,
            "vpnaddr": ""
        }
    ],
    "relativeend": 2147483647,//停止相对时间
    "relativestart": 0,//开始相对时间
    "timestampend": 0,//停止绝对时间
    "timestampstart": 0,//开始绝对时间
    "type": "onlinevideo" //localvideo // picture
}
```

3.9.2 广播背景音乐describe文件信息

代码块

```
{
    //背景音乐列表
    "bgmlist": [
        {
            //如果是在线视频指的是url, 本地文件指的时文
            "mediasource": "C50_home.mp4/",件名
```

```
                "index": 1,//播放顺序
                "showtime": 2147483647,//播放时长
            },
                    {
                "mediasource": "V403_2.mp4",
                "index": 2,
                "showtime": 2147483647,
            }
        ],
        "relativeend": 2147483647,//停止相对时间
        "relativestart": 0,//开始相对时间
        "timestampend": 0,//停止绝对时间
        "timestampstart": 0,//开始绝对时间
        "type": "localbgm" //localbgm
}
```